

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\Ammonia condenser-AK.pvdb

Date of Analysis : Aug 31, 2026 10:06am

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1930.000 mm ID	
Vessel Design Length, Tangent to Tangent	1348.64 cm
Specified Datum Line Distance	0.00 cm
Shell Side Design Temperature	180 °C
Channel Side Design Temperature	85 °C
Shell Side Design Pressure	25.494 kgf/cm ²
Channel Side Design Pressure	10.197 kgf/cm ²
Shop Shell Side Test Pressure	31.540 kgf/cm ²
Field Shell Side Test Pressure	37.232 kgf/cm ²
Shop Channel Side Test Pressure	14.128 kgf/cm ²
Field Channel Side Test Pressure	14.286 kgf/cm ²
Wind Design Code	ASCE/SEI 7-22
Earthquake Design Code	ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Flange	SA-350 LF2	2	...	K03011	+IT
Nozzle	SA-266 2	...	...	K03506	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-334 6	...	...	K03006	+IT
Tubesheet	SA-350 LF2	2	...	K03011	+IT
Flg Bolting	SA-320 L7	...	<= 65	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Lhs Blind flange	10.412	0.00	10.900	3.0000	N/A	No
LHS Ch Flg	10.390	0.00	10.900	3.0000	N/A	No
LHS Ch Shell	10.391	0.00	24.300	3.0000	N/A	No
Main shell-1	25.497	0.00	38.000	1.5000	N/A	No
RHS Ch shell	10.391	0.00	24.300	3.0000	N/A	No
RHS Ch Flg	10.390	0.00	10.900	3.0000	N/A	No
RHS Blind Flg	10.412	0.00	10.900	3.0000	N/A	No

Liquid Level: 215.00 cm Dens.: 999.542 kg/m³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	10.8	10.5	113.0	105.0	101.8	...	1.00	1.00
Body Flg	25.8	15.0	118.0	110.0	104.7	...	1.00	1.00
Cylinder	175.8	150.0	20.0	20.0	10.2	...	1.00	1.00
Cylinder	1137.3	926.0	28.0	28.0	19.2	...	1.00	1.00



Cylinder	1287.3	150.0	20.0	20.0	10.2	...	1.00	1.00
Body Flg	1338.1	15.0	118.0	110.0	104.7	...	1.00	1.00
Body Flg	1348.6	10.5	113.0	105.0	101.8	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
A1-Tube Inlet	WRC-297	0.416	Shell	Passed
B1-Tube Inlet	WRC-297	0.424	Shell	Passed
A2-Shell Inlet	WRC-297	0.669	Shell	Passed
A3-Shell Inlet	WRC-297	0.669	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	275.000	mm
Saddle Bearing Angle	120.000	deg.
Centerline Dimension	1500.000	mm
Wear Pad Width	550.000	mm
Wear Pad Thickness	14.000	mm
Wear Pad Bearing Angle	132.000	deg.
Distance from Saddle to Tangent	1850.000	mm
Baseplate Length	1800.000	mm
Baseplate Thickness	24.000	mm
Baseplate Width	300.000	mm
Number of Ribs (including outside ribs)	4	
Rib Thickness	14.000	mm
Web Thickness	14.000	mm
Height of Center Web	306.000	mm
Number of Bolts in Baseplate	4	

Baseplate Sketch

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	68095.03	kgf
Maximum Transverse Saddle Shear Load	4880.13	kgf
Maximum Longitudinal Saddle Shear Load	9760.26	kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	62308.09	kgf
Maximum Transverse Saddle Shear Load	456.15	kgf
Maximum Longitudinal Saddle Shear Load	262.07	kgf

Weights:

Fabricated - Bare w/o removable internals	85987.5	kg
Shop Test - Fabricated + water (full)	116441.9	kg
Shipping - Fab. + rem. intls.+ shipping app.	85987.5	kg
Erected - Fab. + rem. intls.+ insul. (etc)	85987.5	kg
Empty - Fab. + intls. + details + wghts.	85987.5	kg
Operating - Empty + operating liquid (no ca)	102113.2	kg
Field Test - Empty weight + test medium (full)	116441.9	kg



Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
A1-Tube Inlet	17	...	...	OK	10.05	OK	Passed
B1-Tube Inlet	16.8	...	...	OK	10.18	OK	Passed
A2-Shell Inlet	36.7	...	...	OK	9.60	OK	Passed
A3-Shell Inlet	36.7	...	...	OK	9.60	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 36.7 Nozzle : A2-Shell Inlet [Shellside]
 Minimum MAWP Nozzles : 16.8 Nozzle : B1-Tube Inlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	A1-Tube Inlet	1008.175	90.000	1161.401
30	B1-Tube Inlet	1008.175	270.000	1161.401
40	A2-Shell Inlet	2913.176	360.000	491.851
40	A3-Shell Inlet	10813.177	360.000	491.851

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
A2-Shell Inlet	250 mm	80	WNF	273.0	15.062	...	...	343	300
A3-Shell Inlet	250 mm	80	WNF	273.0	15.062	...	...	343	300
A1-Tube Inlet	600 mm	40	WNF	609.6	17.450	...	...	380	300
B1-Tube Inlet	600 mm	40	WNF	609.6	17.450	...	...	380	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
A2-Shell In	SA-266 2	24.000	10.000	...	...	...
A3-Shell In	SA-266 2	24.000	10.000	...	...	...
A1-Tube Inl	SA-266 2	14.000	10.000	...	...	...
B1-Tube Inl	SA-266 2	14.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
A2-Shell Inlet	291.318	360.0	300.00	0.00	Main shell-1
A3-Shell Inlet	1081.318	360.0	300.00	0.00	Main shell-1
A1-Tube Inlet	100.817	90.0	300.00	0.00	LHS Ch Shell
B1-Tube Inlet	100.817	270.0	300.00	0.00	LHS Ch Shell

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	BLIND FLANGE: 113.0mm THK X 2150.0mm OD	SA-350 LF2
2	WELD NECK FLANGE: 118.0mm THK X 2150.0mm OD	SA-350 LF2
1	CYLINDER: 20.0mm THK X 1930.0mm ID X 150.0cm	SA-516 70
21	BAFFLES: 1892.0mm DIA X 10.0mm THK	SA-516 70
12	TIE RODS: 19mm DIA X 9970mm	SA-193 B7M
2	TUBESHEET: 1986.0mm X 355.0mm THK	SA-350 LF2
4722	TUBES: 9970.0mm X 19.0mm DIA X 2.1mm THK	SA-334 6
2	SADDLE: 275.0mm X 120 DEG	SA-516 70
2	GASKET: 2000.0mm OD X 1962.0mm ID	...
104	BODY FLANGE BOLTS: 27.0mm DIA	SA-320 L7
208	NUTS FOR BODY FLANGE BOLTS: 27.0mm DIA	...
1	CYLINDRICAL SEGMENT 28.0mm THK X 3048.0mm X 926.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 3015.2mm X 926.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm THK X 3048.4mm X 150.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm THK X 3014.7mm X 150.0cm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Main shell-1 [8]		B	2	-18		0.635	28.000	1.00	Yes
RHS Ch shell [8]		B	-7	-8	-29	0.992	20.000	1.00	Yes
Left Saddle [24]		B	-35	-48		0.635	14.000	1.00	Yes
New Saddle [24]		B	-35	-48		0.635	14.000	1.00	Yes
A2-Shell Inlet[1]		B	2	-16		0.668	28.000	1.00	Yes
Nozzle Flg [5]		A	-18	-48					
A3-Shell Inlet[1]		B	2	-16		0.668	28.000	1.00	Yes
Nozzle Flg [5]		A	-18	-48					
Tubesheet: SS[13]		B	12	-104		0.349	88.750	1.00	Yes
Warmest MDMT:			12	-8					
Lhs Blind fla[11]	!		-18	-18	-29	0.974	26.250	1.00	No
LHS Ch Flg [11]	!		-18	-18	-29	0.444	20.000	1.00	No
RHS Ch Flg [11]	!		-18	-18	-29	0.444	20.000	1.00	No
RHS Blind Flg[11]	!		-18	-18	-29	0.974	26.250	1.00	No
LHS Ch Shell [8]		B	-7	-48	-29	0.442	20.000	1.00	No
A1-Tube Inlet [1]		B	-7	-48	-29	0.414	20.000	1.00	No
Nozzle Flg [5]		A	-18	-104					
B1-Tube Inlet [1]		B	-7	-48	-29	0.422	20.000	1.00	No
Nozzle Flg [5]		A	-18	-104					
Tubesheet: CS[14]		B	29	-104		0.084	88.750	1.00	No
Bolting [21]			-101						
Bolting [21]			-101						
Warmest MDMT:			29	-18					
Exchanger Side		Computed MDMT		Required MDMT		Pass/Fail			
Shell		-16.0		-35.0		Fail			
Channel/Tube		-29.0		-35.0		Fail			

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.



UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (Kgfc/m2):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	25.493	kgf/cm ²
Channel Side Design Pressure	10.197	kgf/cm ²
Shell Side Design Temperature	180.0	°C
Channel Side Design Temperature	85.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Lethal	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-35.0	°C
MDMT (CET), Tube Side	-35.0	°C
User defined MAWP, Shell Side	0	kgf/cm ²
User defined MAWP, Channel Side	0	kgf/cm ²
User defined MAPnc, Shell Side	0	kgf/cm ²
User defined MAPnc, Channel Side	0	kgf/cm ²
User defined Test Pres., Shell Side	0	kgf/cm ²
User defined Test Pres., Channel Side	0	kgf/cm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	0 cm
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment	H or Hh 0 cm
Distance Upwind of Crest	Lh 0 cm
Distance from Crest to the Vessel	x 0 cm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 cm

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR] 0.000
Component strength factor	[Rpo] 0.000
Overstrength factor	[Omega0] 0.000
Response Acceleration Factor	[SMS] 0.326
Response Acceleration Factor	[SM1] 0.139
Building Importance Factor	[Ie] 1.000
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
--	-----



Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Flange
Description	Lhs Blind flange
Distance "FROM" to "TO"	10.5 cm
Flange Outside Diameter	2150 mm
Element Thickness	105 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	113 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	10.197 kgf/cm ²
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +IT
Allowable Stress, Ambient	1407.1 kgf/cm ²
Allowable Stress, Operating	1407.1 kgf/cm ²
Allowable Stress, Hydrotest	2294.2 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	2340.9 kgf/cm ²
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K03011
Class / Thickness / Grade	2 ::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	Yes
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	215 cm
Liquid Density	999.54 kg/m ³

Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Flange
Description	LHS Ch Flg
Distance "FROM" to "TO"	15 cm
Flange Inside Diameter	1930 mm
Element Thickness	110 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	118 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	10.197 kgf/cm ²
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 kgf/cm ²



Design Temperature External Pressure 85 °C
Effective Diameter Multiplier 1.2
Material Name SA-350 LF2 +IT
Perform Flange Stress Calculation (Y/N) Y
Weld is pre-Heated No

Element From Node 20
Detail Type Liquid
Detail ID Liquid: 20
Dist. from "FROM" Node / Offset dist 0 cm
Height/Length of Liquid 193 cm
Liquid Density 999.54 kg/m³

Element# 3/7
Element From Node 30
Element To Node 40
Element Type Cylinder
Description LHS Ch Shell
Distance "FROM" to "TO" 150 cm
Inside Diameter 1930 mm
Element Thickness 20 mm
Internal Corrosion Allowance 3 mm
Nominal Thickness 20 mm
External Corrosion Allowance 0 mm
Design Internal Pressure 10.197 kgf/cm²
Design Temperature Internal Pressure 85 °C
Design External Pressure 0 kgf/cm²
Design Temperature External Pressure 85 °C
Effective Diameter Multiplier 1.2
Material Name SA-516 70
Allowable Stress, Ambient 1407.1 kgf/cm²
Allowable Stress, Operating 1407.1 kgf/cm²
Allowable Stress, Hydrotest 2386 kgf/cm²
Material Density 0.00775 kg/cm³
P Number Thickness 32.004 mm
Yield Stress, Operating 2467.6 kgf/cm²
UCS-66 Chart Curve Designation B
External Pressure Chart Name CS-2
UNS Number K02700
Product Form Plate
Efficiency, Longitudinal Seam 1.0
Efficiency, Circumferential Seam 1.0
Weld is pre-Heated No

Element From Node 30
Detail Type Liquid
Detail ID Liquid: 30
Dist. from "FROM" Node / Offset dist 0 cm
Height/Length of Liquid 193 cm
Liquid Density 999.54 kg/m³

Element From Node 30
Detail Type Nozzle
Detail ID A1-Tube Inlet
Dist. from "FROM" Node / Offset dist 75 cm
Nozzle Diameter 600 mm
Nozzle Schedule 40
Nozzle Class 300
Layout Angle 90.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 401.32 kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-266 2

Element From Node 30
Detail Type Nozzle
Detail ID B1-Tube Inlet
Dist. from "FROM" Node / Offset dist 75 cm



Nozzle Diameter	600	mm
Nozzle Schedule	40	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	401.32	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element#	4/7	
Element From Node	40	
Element To Node	60	
Element Type	Cylinder	
Description	Main shell-1	
Distance "FROM" to "TO"	926	cm
Inside Diameter	1930	mm
Element Thickness	28	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	28	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	25.494	kgf/cm ²
Design Temperature Internal Pressure	180	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	180	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	Yes	

Element From Node	40	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	185	cm
Width of Saddle	275	mm
Height of Saddle at Bottom	1500	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	550	mm
Thickness of Wear Plate	14	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm

Element From Node	40	
Detail Type	Saddle	
Detail ID	New Saddle	
Dist. from "FROM" Node / Offset dist	741	cm
Width of Saddle	275	mm
Height of Saddle at Bottom	1500	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	550	mm
Thickness of Wear Plate	14	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	7.9999998e-002	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm

Element From Node	40	
Detail Type	Liquid	



Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	193	cm
Liquid Density	19.991	kg/m ³

Element From Node	40	
Detail Type	Nozzle	
Detail ID	A2-Shell Inlet	
Dist. from "FROM" Node / Offset dist	80	cm
Nozzle Diameter	250	mm
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	360.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	112.98	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	A3-Shell Inlet	
Dist. from "FROM" Node / Offset dist	870	cm
Nozzle Diameter	250	mm
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	360.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	115.87	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element#	5/7	
Element From Node	60	
Element To Node	70	
Element Type	Cylinder	
Description	RHS Ch shell	
Distance "FROM" to "TO"	150	cm
Inside Diameter	1930	mm
Element Thickness	20	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	20	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	10.197	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	193	cm
Liquid Density	999.54	kg/m ³

Element#	6/7	
Element From Node	70	
Element To Node	80	
Element Type	Flange	
Description	RHS Ch Flg	
Distance "FROM" to "TO"	15	cm
Flange Inside Diameter	1930	mm



Element Thickness	110	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	118	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	10.197	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-350 LF2	+IT
Allowable Stress, Ambient	1407.1	kgf/cm ²
Allowable Stress, Operating	1407.1	kgf/cm ²
Allowable Stress, Hydrotest	2294.2	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2340.9	kgf/cm ²
UCS-66 Chart Curve Designation	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K03011	
Class / Thickness / Grade	2 ::	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	193	cm
Liquid Density	999.54	kg/m ³

Element#	7/7
Element From Node	80
Element To Node	90
Element Type	Flange
Description	RHS Blind Flg
Distance "FROM" to "TO"	10.5 cm
Flange Outside Diameter	2150 mm
Element Thickness	105 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	113 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	10.197 kgf/cm ²
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +IT
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	80
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	215 cm
Liquid Density	999.54 kg/m ³



XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
Lhs Blind flang		10.8175	...	10.5	...
LHS Ch Flg		25.8175	...	15	...
LHS Ch Shell		175.817	...	150	...
Main shell-1		1137.32	...	926	...
RHS Ch shell		1287.32	...	150	...
RHS Ch Flg		1338.14	...	15	...
RHS Blind Flg		1348.64	...	10.5	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flange Input Data Values

Description: LHS Blind Fl :

Lhs Blind flange

Flange Type		Blind	
Design Pressure	P	10.41	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	2150.000	mm
Flange Thickness	t	105.0000	mm
Flange Material		SA-350 LF2 +IT	
Flange Material UNS number		K03011	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-320 L7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Perimeter along the Center of the Bolts	L	6503.097	mm
Diameter of Bolt Circle	C	2070.000	mm
Nominal Bolt Diameter	a	27.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		52	
Flange Face Outside Diameter	Fod	2011.000	mm
Flange Face Inside Diameter	Fid	1930.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2000.000	mm
Gasket Inside Diameter	Gi	1962.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm

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Gasket Contact Width,	$N = (Go - Gi)/2$	19.000 mm
Basic Gasket Width,	$bo = N/2$	9.500 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	7.767 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1984.466 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 1984^2 \cdot 10.4$
 $= 322050.281 \text{ kgf}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:
 $= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$
 $= 2 \cdot 7.77 \cdot \pi \cdot 1984 \cdot 3 \cdot 10.4$
 $= 30251.012 \text{ kgf}$

Design Bolt Load for the Operating Condition [Wo]:
 $= \max(Wo,1 + Wo,2 + H'p, 0)$
 $= \max(322050 + 30251 + 0, 0)$
 $= 352301.312 \text{ kgf}$

Design Bolt Load for the Gasket Seating Condition [Wg]:



$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{\text{Part}} \cdot b_{\text{Part}} \cdot l_p \\
 &= 703 \cdot 7.77 \cdot \pi \cdot 1984 + 0 \cdot 0 \cdot 0 \\
 &= 340439.750 \text{ kgf}
 \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned}
 &= \max((W_o + F_a + 4M_e/G) / S_b, W_g / S_a) \\
 &= \max((352301 + 0 + 4 \cdot 0 / 1984) / 1754, 340440 / 1754) \\
 &= 200.877 \text{ cm}^2
 \end{aligned}$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	200.877	215.203	
Radial Distance between Bolts and Edge:	30.500	40.000	
Circ. Spacing between the Bolts:	65.000	124.983	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\
 &= 215 \cdot 1754 / (703 \cdot \pi (2000 + 1962)) \\
 &= 4.313 \text{ mm}
 \end{aligned}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= S_a (A_m + A_b) / 2 \\
 &= 1754 (201 + 215) / 2 \\
 &= 364864.03 \text{ kgf}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= W_o \\
 &= 352301.31 \text{ kgf}
 \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned}
 &= (C - G) / 2 \\
 &= (2070 - 1984) / 2 \\
 &= 42.7669 \text{ mm}
 \end{aligned}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$\begin{aligned}
 &= 1.9 \cdot W_{m1} \cdot h_G \cdot B_{sc} / (t^2 \cdot G) + C \cdot Z \cdot P_{eq} \cdot G^2 / t^2 \\
 &= 1.9 \cdot 352301 \cdot 42.8 \cdot 1 / (102^2 \cdot 1984) + 0.3 \cdot 1 \cdot 10.4 \cdot 1984^2 / 102^2 \\
 &= 1321.03 \text{ kgf/cm}^2
 \end{aligned}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$\begin{aligned}
 &= 1.9 \cdot W \cdot h_G \cdot B_{sc} / (t^2 \cdot G) \\
 &= 1.9 \cdot 364864 \cdot 42.8 \cdot 1 / (102^2 \cdot 1984) \\
 &= 143.60 \text{ kgf/cm}^2
 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned}
 &= W_{m1} / A_b \\
 &= 352301 / 215 \\
 &= 1637.07 \text{ kgf/cm}^2
 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned}
 &= (W_{m2} / A_b) \\
 &= (340440 / 215) \\
 &= 1581.95 \text{ kgf/cm}^2
 \end{aligned}$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Tangential Flange	1321.03	1407.13	143.60	1407.13
Bolting	1637.07	1753.82	1581.95	1753.82

Reqd. Blind Flange Thickness at Center 101.830 mm



Reqd. Blind Flange Thickness at Gasket 35.584 mm
Estimated M.A.W.P. (Operating) 11.1 kgf/cm²
Estimated Finished Weight of Flange at given Thk. 2954.5 kg
Estimated Unfinished Weight of Forging at given Thk 2954.5 kg

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -18 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Tangential Flange	1246.62	1407.13	135.51	1407.13
Bolting	1637.07	1753.82	1581.95	1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 98.831 mm

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Flange Input Data Values

Description: LHS Ch Flg :

LHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	10.39	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1930.000	mm
Flange Outside Diameter	A	2150.000	mm
Flange Thickness	t	110.0000	mm
Thickness of Hub at Small End	go	20.0000	mm
Thickness of Hub at Large End	g1	30.0000	mm
Length of Hub	h	40.0000	mm
Flange Material		SA-350 LF2 +IT	
Flange Material UNS number		K03011	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-320 L7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	2070.000	mm
Nominal Bolt Diameter	a	27.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		52	
Flange Face Outside Diameter	Fod	2011.000	mm
Flange Face Inside Diameter	Fid	1930.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2000.000	mm
Gasket Inside Diameter	Gi	1962.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (10.4 \cdot (1930/2 + 3)) / (1407 \cdot 1 - 0.6 \cdot 10.4) + Ca$$

$$= 10.1795 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 17) / (968 + 0.6 \cdot 17)$$

$$= 24.454 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 20) / (965 + 0.6 \cdot 20)$$

$$= 28.805 \text{ kgf/cm}^2$$

Corroded Flange Thickness, tc	= T-ci	107.000 mm
Corroded Flange ID, Bcor	= B+2*ci	1936.000 mm
Corroded Large Hub, g1Cor	= g1-ci	27.000 mm
Corroded Small Hub, g0Cor	= go-ci	17.000 mm
Code R Dimension, R	= ((C-Bcor)/2)-g1cor	40.000 mm
Gasket Contact Width, N	= (Go - Gi)/2	19.000 mm



Basic Gasket Width, $bo = N/2$ 9.500 mm
 Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$ 7.767 mm
 Gasket Reaction Diameter, $G = Go - 2 \cdot b$ 1984.466 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 1984^2 \cdot 10.4$
 $= 321370.125 \text{ kgf}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:
 $= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$
 $= 2 \cdot 7.77 \cdot \pi \cdot 1984 \cdot 3 \cdot 10.4$
 $= 30187.121 \text{ kgf}$

Hydrostatic End Load at Flange ID [Hd]:
 $= \pi/4 \cdot Bcor^2 \cdot P$
 $= \pi/4 \cdot 1936^2 \cdot 10.4$
 $= 305864.281 \text{ kgf}$

Pressure Force on Flange Face [Ht]:
 $= H - Hd$
 $= 321370 - 305864$
 $= 15505.844 \text{ kgf}$

Design Bolt Load for the Operating Condition [Wo]:
 $= \max(Wo,1 + Wo,2 + H'p, 0)$
 $= \max(321370 + 30187 + 0, 0)$
 $= 351557.250 \text{ kgf}$

Design Bolt Load for the Gasket Seating Condition [Wg]:
 $= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$
 $= 703 \cdot 7.77 \cdot \pi \cdot 1984 + 0 \cdot 0 \cdot 0$
 $= 340439.750 \text{ kgf}$

Required Bolt Area [Am]:
 $= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$
 $= \max((351557 + 0 + 4 \cdot 0/1984)/1754, 340440/1754)$
 $= 200.453 \text{ cm}^2$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:
 $= 2a + 6t/(m + 0.5)$
 $= 2 \cdot 27 + 6 \cdot 107/(3 + 0.5)$
 $= 237.429 \text{ mm}$

Actual Circumferential Bolt Spacing [Bs]:
 $= C \cdot \sin(\pi / n)$
 $= 2070 \cdot \sin(\pi/52)$
 $= 124.983 \text{ mm}$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:
 $= \max(\sqrt{ Bs/(2a + t) }, 1)$
 $= \max(\sqrt{ 125/(2 \cdot 27 + 107) }, 1)$
 $= 1.0000$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	200.453	215.203	
Radial Distance between Hub and Bolts:	38.000	40.000	
Radial Distance between Bolts and Edge:	30.500	40.000	
Circ. Spacing between the Bolts:	65.000	124.983	237.429

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
 $= Ab \cdot Sa/(y \cdot \pi(Go + Gi))$
 $= 215 \cdot 1754/(703 \cdot \pi(2000 + 1962))$
 $= 4.313 \text{ mm}$



Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 1754 (200 + 215) / 2$$

$$= 364492.00 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 351557 - 321370$$

$$= 30187.12 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2070 - 1984) / 2$$

$$= 42.7669 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((2070 - 1936) / 2 + 42.8)$$

$$= 54.8834 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2070 - 1936 - 27) / 2$$

$$= 53.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	305864.	53.5000	1.0000	16364.
Face Pressure, Mt	15506.	54.8834	1.0000	851.
Gasket Load, Mg	30187.	42.7669	1.0000	1291.
Gasket Seating, Matm	364492.	42.7669	1.0000	15588.

Total Moment for Operation, Mo 18506. kgf-m.
 Total Moment for Gasket seating, Mg 15588. kgf-m.

Effective Hub Length, $ho = \sqrt{Bcor \cdot goCor}$ 181.417 mm
 Hub Ratio, $h/h0 = HL / H0$ 0.220
 Thickness Ratio, $g1/g0 = (g1Cor/goCor)$ 1.588

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.895
 Factor V per 4.16.5 0.390
 Factor f per 4.16.5 1.601

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 2150 / 1936$$

$$= 1.111$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K)) / (K^2 - 1))) / (K - 1)$$

$$= (0.66845 + (5.7169(1.11^2 \cdot \log_{10}(1.11)) / (1.11^2 - 1))) / (1.11 - 1)$$

$$= 18.497$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.11^2(1 + 8.55246 \cdot \log_{10}(1.11)) - 1) / ((1.0472 + 1.9448 \cdot 1.11^2) \cdot (1.11 - 1))$$

$$= 1.873$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$



$$= (1.11^2(1 + 8.55246 \cdot \log_{10}(1.11)) - 1) / (1.36136(1.11^2 - 1) \cdot (1.11 - 1))$$
$$= 20.326$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$
$$= (1.11^2 + 1) / (1.11^2 - 1)$$
$$= 9.573$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$
$$= 20.3 \cdot 17^2 \cdot 181 / 0.39$$
$$= 2731361.250 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$
$$= 0.89 / 181$$
$$= 0.005 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3 / d$$
$$= (107 \cdot 0.0049 + 1) / 1.87 + 107^3 / 2731361$$
$$= 1.264$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$
$$= (1.6 \cdot 18506 / 1936) / (1.26 \cdot 27^2)$$
$$= 1660.26 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$
$$= (1.6 \cdot 15588 / 1936) / (1.26 \cdot 27^2)$$
$$= 1398.51 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$
$$= (1.7 \cdot 18506 / 1936) / (1.26 \cdot 107^2)$$
$$= 112.54 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$
$$= (1.7 \cdot 15588 / 1936) / (1.26 \cdot 107^2)$$
$$= 94.80 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$
$$= (18.5 \cdot 18506 / (107^2 \cdot 1936)) - 9.57 \cdot 113$$
$$= 466.97 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$
$$= (18.5 \cdot 15588 / (107^2 \cdot 1936)) - 9.57 \cdot 94.8$$
$$= 393.34 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$
$$= (1660 + \max(113, 467)) / 2$$
$$= 1063.61 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$
$$= (1399 + \max(94.8, 393)) / 2$$
$$= 895.93 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$
$$= 351557 / 215$$
$$= 1633.61 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm^2 / Ab)$$

$$= (340440/215)$$

$$= 1581.95 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	1660.26	2110.70	1398.51	2110.70
Radial Flange	112.54	1407.13	94.80	1407.13
Tangential Flange	466.97	1407.13	393.34	1407.13
Maximum Average	1063.61	1407.13	895.93	1407.13
Bolting	1633.61	1753.82	1581.95	1753.82

Minimum Required Flange Thickness [Rigidity] 104.699 mm

Estimated M.A.W.P. (Operating) 11.1 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 648.6 kg
 Estimated Unfinished Weight of Forging at given Thk 819.6 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 15588 / 1 \cdot 100000 \cdot 0.39 / (1.19 \cdot 2052964 \cdot 17^2 \cdot 181 \cdot 0.3)$
 $= 0.827$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 18506 / 1 \cdot 100000 \cdot 0.39 / (1.19 \cdot 2015652 \cdot 17^2 \cdot 181 \cdot 0.3)$
 $= 1.000$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 15588 / 1 \cdot 100000 \cdot 0.39 / (1.26 \cdot 2052964 \cdot 17^2 \cdot 181 \cdot 0.3)$
 $= 0.776$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 18506 / 1 \cdot 100000 \cdot 0.39 / (1.26 \cdot 2015652 \cdot 17^2 \cdot 181 \cdot 0.3)$
 $= 0.939$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -18 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	1408.47	2110.70	1156.13	2110.70
Radial Flange	117.62	1407.13	96.54	1407.13
Tangential Flange	367.73	1407.13	301.85	1407.13
Maximum Average	888.10	1407.13	728.99	1407.13
Bolting	1633.61	1753.82	1581.95	1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 76.733 mm



Flange Input Data Values

Description: RHS Ch Flg :

RHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	10.39	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1930.000	mm
Flange Outside Diameter	A	2150.000	mm
Flange Thickness	t	110.0000	mm
Thickness of Hub at Small End	go	20.0000	mm
Thickness of Hub at Large End	g1	30.0000	mm
Length of Hub	h	40.0000	mm
Flange Material		SA-350 LF2 +IT	
Flange Material UNS number		K03011	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-320 L7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	2070.000	mm
Nominal Bolt Diameter	a	27.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		52	
Flange Face Outside Diameter	Fod	2011.000	mm
Flange Face Inside Diameter	Fid	1930.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2000.000	mm
Gasket Inside Diameter	Gi	1962.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (10.4 \cdot (1930/2 + 3)) / (1407 \cdot 1 - 0.6 \cdot 10.4) + Ca$$

$$= 10.1795 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 17) / (968 + 0.6 \cdot 17)$$

$$= 24.454 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 20) / (965 + 0.6 \cdot 20)$$

$$= 28.805 \text{ kgf/cm}^2$$

Corroded Flange Thickness, tc = T-ci	107.000	mm
Corroded Flange ID, Bcor = B+2*ci	1936.000	mm
Corroded Large Hub, g1Cor = g1-ci	27.000	mm
Corroded Small Hub, g0Cor = go-ci	17.000	mm
Code R Dimension, R = ((C-Bcor)/2)-g1cor	40.000	mm
Gasket Contact Width, N = (Go - Gi)/2	19.000	mm



Basic Gasket Width, $bo = N/2$ 9.500 mm
 Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$ 7.767 mm
 Gasket Reaction Diameter, $G = Go - 2 \cdot b$ 1984.466 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 1984^2 \cdot 10.4$
 $= 321370.125 \text{ kgf}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:
 $= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$
 $= 2 \cdot 7.77 \cdot \pi \cdot 1984 \cdot 3 \cdot 10.4$
 $= 30187.121 \text{ kgf}$

Hydrostatic End Load at Flange ID [Hd]:
 $= \pi/4 \cdot Bcor^2 \cdot P$
 $= \pi/4 \cdot 1936^2 \cdot 10.4$
 $= 305864.281 \text{ kgf}$

Pressure Force on Flange Face [Ht]:
 $= H - Hd$
 $= 321370 - 305864$
 $= 15505.844 \text{ kgf}$

Design Bolt Load for the Operating Condition [Wo]:
 $= \max(Wo,1 + Wo,2 + H'p, 0)$
 $= \max(321370 + 30187 + 0, 0)$
 $= 351557.250 \text{ kgf}$

Design Bolt Load for the Gasket Seating Condition [Wg]:
 $= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$
 $= 703 \cdot 7.77 \cdot \pi \cdot 1984 + 0 \cdot 0 \cdot 0$
 $= 340439.750 \text{ kgf}$

Required Bolt Area [Am]:
 $= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$
 $= \max((351557 + 0 + 4 \cdot 0/1984)/1754, 340440/1754)$
 $= 200.453 \text{ cm}^2$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:
 $= 2a + 6t/(m + 0.5)$
 $= 2 \cdot 27 + 6 \cdot 107/(3 + 0.5)$
 $= 237.429 \text{ mm}$

Actual Circumferential Bolt Spacing [Bs]:
 $= C \cdot \sin(\pi / n)$
 $= 2070 \cdot \sin(\pi/52)$
 $= 124.983 \text{ mm}$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:
 $= \max(\sqrt{ Bs/(2a + t) }, 1)$
 $= \max(\sqrt{ 125/(2 \cdot 27 + 107) }, 1)$
 $= 1.0000$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	200.453	215.203	
Radial Distance between Hub and Bolts:	38.000	40.000	
Radial Distance between Bolts and Edge:	30.500	40.000	
Circ. Spacing between the Bolts:	65.000	124.983	237.429

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
 $= Ab \cdot Sa/(y \cdot \pi(Go + Gi))$
 $= 215 \cdot 1754/(703 \cdot \pi(2000 + 1962))$
 $= 4.313 \text{ mm}$



Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 1754 (200 + 215) / 2$$

$$= 364492.00 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 351557 - 321370$$

$$= 30187.12 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2070 - 1984) / 2$$

$$= 42.7669 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((2070 - 1936) / 2 + 42.8)$$

$$= 54.8834 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2070 - 1936 - 27) / 2$$

$$= 53.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	305864.	53.5000	1.0000	16364.
Face Pressure, Mt	15506.	54.8834	1.0000	851.
Gasket Load, Mg	30187.	42.7669	1.0000	1291.
Gasket Seating, Matm	364492.	42.7669	1.0000	15588.

Total Moment for Operation, Mo 18506. kgf-m.
 Total Moment for Gasket seating, Mg 15588. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 181.417 mm
 Hub Ratio, h/h0 = HL / H0 0.220
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.588

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.895
 Factor V per 4.16.5 0.390
 Factor f per 4.16.5 1.601

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 2150 / 1936$$

$$= 1.111$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K)) / (K^2 - 1))) / (K - 1)$$

$$= (0.66845 + (5.7169(1.11^2 \cdot \log_{10}(1.11)) / (1.11^2 - 1))) / (1.11 - 1)$$

$$= 18.497$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.11^2(1 + 8.55246 \cdot \log_{10}(1.11)) - 1) / ((1.0472 + 1.9448 \cdot 1.11^2) \cdot (1.11 - 1))$$

$$= 1.873$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$



$$= (1.11^2(1 + 8.55246 \cdot \log_{10}(1.11)) - 1) / (1.36136(1.11^2 - 1) \cdot (1.11 - 1)) \\ = 20.326$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1) \\ = (1.11^2 + 1) / (1.11^2 - 1) \\ = 9.573$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V \\ = 20.3 \cdot 17^2 \cdot 181 / 0.39 \\ = 2731361.250 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0 \\ = 0.89 / 181 \\ = 0.005 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3 / d \\ = (107 \cdot 0.0049 + 1) / 1.87 + 107^3 / 2731361 \\ = 1.264$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ = (1.6 \cdot 18506 / 1936) / (1.26 \cdot 27^2) \\ = 1660.26 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ = (1.6 \cdot 15588 / 1936) / (1.26 \cdot 27^2) \\ = 1398.51 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ = (1.7 \cdot 18506 / 1936) / (1.26 \cdot 107^2) \\ = 112.54 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ = (1.7 \cdot 15588 / 1936) / (1.26 \cdot 107^2) \\ = 94.80 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ = (18.5 \cdot 18506 / (107^2 \cdot 1936)) - 9.57 \cdot 113 \\ = 466.97 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\ = (18.5 \cdot 15588 / (107^2 \cdot 1936)) - 9.57 \cdot 94.8 \\ = 393.34 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2 \\ = (1660 + \max(113, 467)) / 2 \\ = 1063.61 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2 \\ = (1399 + \max(94.8, 393)) / 2 \\ = 895.93 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab \\ = 351557 / 215 \\ = 1633.61 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm^2 / Ab)$$

$$= (340440/215)$$

$$= 1581.95 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	1660.26	2110.70	1398.51	2110.70
Radial Flange	112.54	1407.13	94.80	1407.13
Tangential Flange	466.97	1407.13	393.34	1407.13
Maximum Average	1063.61	1407.13	895.93	1407.13
Bolting	1633.61	1753.82	1581.95	1753.82

Minimum Required Flange Thickness [Rigidity] 104.699 mm

Estimated M.A.W.P. (Operating) 11.1 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 648.6 kg
 Estimated Unfinished Weight of Forging at given Thk 819.6 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 15588 / 1 \cdot 100000 \cdot 0.39 / (1.19 \cdot 2052964 \cdot 17^2 \cdot 181 \cdot 0.3)$
 $= 0.827$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 18506 / 1 \cdot 100000 \cdot 0.39 / (1.19 \cdot 2015652 \cdot 17^2 \cdot 181 \cdot 0.3)$
 $= 1.000$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 15588 / 1 \cdot 100000 \cdot 0.39 / (1.26 \cdot 2052964 \cdot 17^2 \cdot 181 \cdot 0.3)$
 $= 0.776$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 18506 / 1 \cdot 100000 \cdot 0.39 / (1.26 \cdot 2015652 \cdot 17^2 \cdot 181 \cdot 0.3)$
 $= 0.939$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -18 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	1408.47	2110.70	1156.13	2110.70
Radial Flange	117.62	1407.13	96.54	1407.13
Tangential Flange	367.73	1407.13	301.85	1407.13
Maximum Average	888.10	1407.13	728.99	1407.13
Bolting	1633.61	1753.82	1581.95	1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 76.733 mm



Flange Input Data Values

Description: RHS Blind Fl :

RHS Blind Flg

Flange Type		Blind	
Design Pressure	P	10.41	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	2150.000	mm
Flange Thickness	t	105.0000	mm
Flange Material		SA-350 LF2 +IT	
Flange Material UNS number		K03011	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-320 L7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Perimeter along the Center of the Bolts	L	6503.097	mm
Diameter of Bolt Circle	C	2070.000	mm
Nominal Bolt Diameter	a	27.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		52	
Flange Face Outside Diameter	Fod	2011.000	mm
Flange Face Inside Diameter	Fid	1930.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2000.000	mm
Gasket Inside Diameter	Gi	1962.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm

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Gasket Contact Width,	$N = (Go - Gi)/2$	19.000 mm
Basic Gasket Width,	$bo = N/2$	9.500 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	7.767 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1984.466 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 1984^2 \cdot 10.4$
 $= 322050.281 \text{ kgf}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:
 $= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$
 $= 2 \cdot 7.77 \cdot \pi \cdot 1984 \cdot 3 \cdot 10.4$
 $= 30251.012 \text{ kgf}$

Design Bolt Load for the Operating Condition [Wo]:
 $= \max(Wo,1 + Wo,2 + H'p, 0)$
 $= \max(322050 + 30251 + 0, 0)$
 $= 352301.312 \text{ kgf}$

Design Bolt Load for the Gasket Seating Condition [Wg]:



$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{\text{Part}} \cdot b_{\text{Part}} \cdot l_p \\
 &= 703 \cdot 7.77 \cdot \pi \cdot 1984 + 0 \cdot 0 \cdot 0 \\
 &= 340439.750 \text{ kgf}
 \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned}
 &= \max((W_o + F_a + 4M_e/G) / S_b, W_g / S_a) \\
 &= \max((352301 + 0 + 4 \cdot 0 / 1984) / 1754, 340440 / 1754) \\
 &= 200.877 \text{ cm}^2
 \end{aligned}$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	200.877	215.203	
Radial Distance between Bolts and Edge:	30.500	40.000	
Circ. Spacing between the Bolts:	65.000	124.983	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\
 &= 215 \cdot 1754 / (703 \cdot \pi (2000 + 1962)) \\
 &= 4.313 \text{ mm}
 \end{aligned}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= S_a (A_m + A_b) / 2 \\
 &= 1754 (201 + 215) / 2 \\
 &= 364864.03 \text{ kgf}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= W_o \\
 &= 352301.31 \text{ kgf}
 \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned}
 &= (C - G) / 2 \\
 &= (2070 - 1984) / 2 \\
 &= 42.7669 \text{ mm}
 \end{aligned}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$\begin{aligned}
 &= 1.9 \cdot W_{m1} \cdot h_G \cdot B_{sc} / (t^2 \cdot G) + C \cdot Z \cdot P_{eq} \cdot G^2 / t^2 \\
 &= 1.9 \cdot 352301 \cdot 42.8 \cdot 1 / (102^2 \cdot 1984) + 0.3 \cdot 1 \cdot 10.4 \cdot 1984^2 / 102^2 \\
 &= 1321.03 \text{ kgf/cm}^2
 \end{aligned}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$\begin{aligned}
 &= 1.9 \cdot W \cdot h_G \cdot B_{sc} / (t^2 \cdot G) \\
 &= 1.9 \cdot 364864 \cdot 42.8 \cdot 1 / (102^2 \cdot 1984) \\
 &= 143.60 \text{ kgf/cm}^2
 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned}
 &= W_{m1} / A_b \\
 &= 352301 / 215 \\
 &= 1637.07 \text{ kgf/cm}^2
 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned}
 &= (W_{m2} / A_b) \\
 &= (340440 / 215) \\
 &= 1581.95 \text{ kgf/cm}^2
 \end{aligned}$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Tangential Flange	1321.03	1407.13	143.60	1407.13
Bolting	1637.07	1753.82	1581.95	1753.82

Reqd. Blind Flange Thickness at Center 101.830 mm



Reqd. Blind Flange Thickness at Gasket 35.584 mm
Estimated M.A.W.P. (Operating) 11.1 kgf/cm²
Estimated Finished Weight of Flange at given Thk. 2954.5 kg
Estimated Unfinished Weight of Forging at given Thk 2954.5 kg

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -18 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Tangential Flange	1246.62	1407.13	135.51	1407.13
Bolting	1637.07	1753.82	1581.95	1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 98.831 mm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
Lhs Blind flange		10.412	113	3	2150	1407.1	1
LHS Ch Flg		10.39	118	3	1930	1407.1	2
LHS Ch Shell		10.391	20	3	1930	1407.1	3
Main shell-1		25.497	28	1.5	1930	1407.1	4
RHS Ch shell		10.391	20	3	1930	1407.1	5
RHS Ch Flg		10.39	118	3	1930	1407.1	6
RHS Blind Flg		10.412	113	3	2150	1407.1	7

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
Lhs Blind flange		10.1974	10.9	11.2	105	101.83	1
LHS Ch Flg		10.1974	10.9	11.2	110	104.699	2
LHS Ch Shell		10.1974	24.3	28.8	20	10.1798	3
Main shell-1		25.4935	38	40.1	28	19.2055	4
RHS Ch shell		10.1974	24.3	28.8	20	10.1798	5
RHS Ch Flg		10.1974	10.9	11.2	110	104.699	6
RHS Blind Flg		10.1974	10.9	11.2	105	101.83	7

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 24.3 kgf/cm²
 Shell Side MAPnc = 28.8 kgf/cm²
 Channel Side MAWP = 10.9 kgf/cm²
 Channel Side MAPnc = 11.2 kgf/cm²

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:					
Design Pressure	P	10.391 kgf/cm²	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	85 °C	
Length		150.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	1930.000 mm	Actual Stress	597.9 kgf/cm²	
Spec. Min. Thk t _e , or s _a		20.0000 mm	Spec. Nominal Thk.	20.000 mm	
Surface Area		92834.1 cm²	SG of Contents	1.000	
Element Volume		4388304.5 cm³	Weight of Contents	4386.3 kgf	
Empty Weight		1424.4 kgf	Operating Weight	5810.7 kgf	
			Comp. All. Ope. Stress	1290.0 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (10.4 \cdot 968) / (1407 \cdot 1 - 0.6 \cdot 10.4)$$

$$= 7.1798 + 3.0000 = 10.1798 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.193 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 17) / (968 + 0.6 \cdot 17)$$

$$= 24.5 - 0.19 = 24.3 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 20) / (965 + 0.6 \cdot 20)$$

$$= 28.8 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (10.4 \cdot (968 + 0.6 \cdot 17)) / (1 \cdot 17)$$

$$= 597.888 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.026 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 20, t_r = 7.51, c = 3 mm, E* = 1

Thickness Ratio = t_r * E* / (t_g - c) = 0.44, Temp. Reduction = 41 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	10.868 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-35 °C
Governing Thk.	20.000 mm	Basic MDMT	-7 °C

PV Elite 28 Licensee: SL User
FileName : Ammonia condenser-AK
Element: LHS Ch Shell

Step: 8 10:06am Aug 31,2026

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Stress/Thk. Ratio	0.442	Temperature Difference	41 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Main shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	25.497 kgf/cm²	Int. Design Temperature	180 °C	
Material		SA-516 70	Ext. Design Temperature	180 °C	
Length		926.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	1930.000 mm	Actual Stress	945.2 kgf/cm²	
Spec. Min. Thk t _e , or s _a		28.0000 mm	Spec. Nominal Thk.	28.000 mm	
Surface Area		577750.3 cm²	SG of Contents	0.020	
Element Volume		14627608.0 cm³	Weight of Contents	292.4 kgf	
Empty Weight		12361.0 kgf	Operating Weight	12653.4 kgf	
			Comp. All. Ope. Stress	1298.4 kgf/cm²	

Radiography Information:			
User Defined			User Defined
Circ. Joint Efficiency	100 %		Long. Joint Eff. E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (25.5 \cdot 966) / (1407 \cdot 1 - 0.6 \cdot 25.5)$$

$$= 17.7055 + 1.5000 = 19.2055 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.004 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 26.5) / (966 + 0.6 \cdot 26.5)$$

$$= 38 - 0.0039 = 38 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 28) / (965 + 0.6 \cdot 28)$$

$$= 40.1 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (25.5 \cdot (966 + 0.6 \cdot 26.5)) / (1 \cdot 26.5)$$

$$= 945.231 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.430 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 28, t_r = 16.8, c = 1.5 mm, E* = 1

Thickness Ratio = t_r * E* / (t_g - c) = 0.64, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C

Min Metal Temp. at Required thickness (UCS 66.1) -18 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:	
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PV Elite 28 Licensee: SL User
FileName : Ammonia condenser-AK
Element: Main shell-1

Step: 9 10:06am Aug 31,2026

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HEXAGON

MDMT Curve	B	Press. at MDMT	24.261 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-35 °C
Governing Thk.	28.000 mm	Basic MDMT	2 °C
Stress/Thk. Ratio	0.635	Temperature Difference	20 °C
UCS-66(c) red.	No	Computed Min. Temp.	-18 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: RHS Ch shell

Cylindrical Shell Design Information:					
Design Pressure	P	10.391 kgf/cm²	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	85 °C	
Length		150.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	1930.000 mm	Actual Stress	597.9 kgf/cm²	
Spec. Min. Thk t _e , or s _a		20.0000 mm	Spec. Nominal Thk.	20.000 mm	
Surface Area		92834.1 cm²	SG of Contents	1.000	
Element Volume		2369482.2 cm³	Weight of Contents	2368.4 kgf	
Empty Weight		1424.4 kgf	Operating Weight	3792.8 kgf	
			Comp. All. Ope. Stress	1298.4 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (10.4 \cdot 968) / (1407 \cdot 1 - 0.6 \cdot 10.4)$$

$$= 7.1798 + 3.0000 = 10.1798 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.193 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 17) / (968 + 0.6 \cdot 17)$$

$$= 24.5 - 0.19 = 24.3 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 20) / (965 + 0.6 \cdot 20)$$

$$= 28.8 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (10.4 \cdot (968 + 0.6 \cdot 17)) / (1 \cdot 17)$$

$$= 597.888 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.026 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 20, t_r = 16.9, c = 3 mm, E* = 1

Thickness Ratio = t_r * E* / (t_g - c) = 0.99, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -8 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:	
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MDMT Curve	B	Press. at MDMT	24.261 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-35 °C
Governing Thk.	20.000 mm	Basic MDMT	-7 °C
Stress/Thk. Ratio	0.992	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-8 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %



Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	31.540 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	13.257 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	37.232 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	26.687 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	14.582 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	14.582 kgf/cm ²
App 27-4	= M.A.W.P.	24.261 kgf/cm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	14.128 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	13.257 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	14.286 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	11.954 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	14.582 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	14.582 kgf/cm ²
App 27-4	= M.A.W.P.	10.868 kgf/cm ²

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	699.6	2386.0	0.293	14.32
Main shell-1	1112.7	2386.0	0.466	31.73
RHS Ch shell	1550.1	2386.0	0.650	31.73

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
A1-Tube Inlet	Nozzle	1407.13	1407.13	1.000
B1-Tube Inlet	Nozzle	1407.13	1407.13	1.000
A2-Shell Inlet	Nozzle	1407.13	1407.13	1.000
A3-Shell Inlet	Nozzle	1407.13	1407.13	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Lhs Blind flange	1407.13	1407.13	1.000
LHS Ch Flg	1407.13	1407.13	1.000
LHS Ch Shell	1407.13	1407.13	1.000
Main shell-1	1407.13	1407.13	1.000
RHS Ch shell	1407.13	1407.13	1.000
RHS Ch Flg	1407.13	1407.13	1.000
RHS Blind Flg	1407.13	1407.13	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
Tube Material	1203.20	1203.20	1.000
Tubesheet Material	1407.13	1407.13	1.000



Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
A1-Tube Inlet	244.42	2294.24	0.107
B1-Tube Inlet	244.42	2294.24	0.107
A2-Shell Inlet	274.93	2294.24	0.120
A3-Shell Inlet	274.93	2294.24	0.120

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	31.540 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	13.257 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	37.232 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	26.687 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	14.582 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	14.582 kgf/cm ²
App 27-4	= M.A.W.P.	24.261 kgf/cm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	14.128 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	13.257 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	14.286 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	11.954 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	14.582 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	14.582 kgf/cm ²
App 27-4	= M.A.W.P.	10.868 kgf/cm ²

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	833.2	2386.0	0.349	14.48
Main shell-1	1387.4	2386.0	0.581	37.42
RHS Ch shell	2153.5	2386.0	0.903	37.42

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
A1-Tube Inlet	Nozzle	1407.13	1407.13	1.000
B1-Tube Inlet	Nozzle	1407.13	1407.13	1.000
A2-Shell Inlet	Nozzle	1407.13	1407.13	1.000
A3-Shell Inlet	Nozzle	1407.13	1407.13	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Lhs Blind flange	1407.13	1407.13	1.000
LHS Ch Flg	1407.13	1407.13	1.000
LHS Ch Shell	1407.13	1407.13	1.000
Main shell-1	1407.13	1407.13	1.000
RHS Ch shell	1407.13	1407.13	1.000
RHS Ch Flg	1407.13	1407.13	1.000
RHS Blind Flg	1407.13	1407.13	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
Tube Material	1203.20	1203.20	1.000
Tubesheet Material	1407.13	1407.13	1.000



Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
A1-Tube Inlet	247.12	2294.24	0.108
B1-Tube Inlet	247.12	2294.24	0.108
A2-Shell Inlet	324.25	2294.24	0.141
A3-Shell Inlet	324.25	2294.24	0.141

Elements Suitable for Test Pressure.

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MDMT Calculations for Stiffening Rings:

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main shell-1

Shell Test Pressure		37.23 kgf/cm ²
Shell Thickness	ts	28.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1930.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	1407.13 kgf/cm ²
Shell Allowable Stress at Ambient		1407.13 kgf/cm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		14.29 kgf/cm ²
Channel Thickness	tc	20.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1930.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	1407.13 kgf/cm ²
Channel Allowable Stress at Ambient		1407.13 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	4722
Tube Wall Thickness	et	2.1000 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	9970.00 mm
Straight Tube Length (bet. inner tubsht faces) L		9260.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-334 6
Tube Material UNS Number		K03006
Is this a Welded Tube		No
Tube Material Specification used	Smls. & wld. tube	
Tube Allowable Stress at Temperature		1203.20 kgf/cm ²
Tube Allowable Stress At Ambient		1203.20 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	2447.19 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	25.4000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.3000 mm
Groove Weld Leg	ag	1.3000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	940.081 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	6.0000 mm
Area of Tube Layout	Ap	26284.4 cm ²
Perimeter of Tube Layout	Cp	6032.4 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A



Tubesheet Design Metal Temperature T 21.11 °C
 Tubesheet Material SA-350 LF2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03011
 Tubesheet Allowable Stress at Temperature S 1407.13 kgf/cm²
 Tubesheet Allowable Stress at Ambient Tt 1407.13 kgf/cm²
 Thickness of Tubesheet h 355.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm
 Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
 Tubesheet Outside Diameter A 1986.000 mm

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 470.000 mm
 Tube End condition corresponding to Span (l) k 1.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes

Mean Shell Metal Temp. along Shell len. Tsm 21.11 °C
 Mean Tube Metal Temp. along Tube length Ttm 21.11 °C
 Junction Stress Reduction option Perform Plastic Calculation

Tubesheet Integral with Both
 Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²



Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (14.3 \cdot (19/2 - 0)) / (1203 \cdot 1 + 0.4 \cdot 14.3)$
 $= 0.1126 + 0.0000 = 0.1126 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 21.11 °C
 Elastic Modulus for Material 2038903.00 kgf/cm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1000	19.05	9260.00	9.07	50.0000	0.0133672	1251.46

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 195 \text{ kgf/cm}^2$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.5960	19.05	9260.00	31.96	50.0000	0.0010767	892.62

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 893) / (3 \cdot 32) = 37.2 \text{ kgf/cm}^2$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.5960 mm
Allowable Internal Pressure at Corroded thickness	290.93 kgf/cm ²
Required Internal Design Pressure	14.29 kgf/cm ²
Allowable External Pressure at Corroded thickness	194.71 kgf/cm ²
Required External Design Pressure	37.23 kgf/cm ²
Required Thickness due to Shell Side pressure	0.5960 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [p]:

$= l_{tx} / h$ (modified for corrosion if present)
 $= 0/355$
 $= 0.0000$ (must be $0 \leq p \leq 1$)

Effective tube hole diameter [d*]:

$= \max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (p); dt - 2 \cdot tt)$
 $= \max(19 - 2 \cdot 2.1 (2067025/2067025) \cdot (1203/1407) \cdot (0); 19 - 2 \cdot 2.1)$
 $= 19.0500 \text{ mm}$

Equivalent outer tube limit circle diameter [Do]:

$= 2 \cdot r_o + dt$
 $= 2 \cdot 940 + 19$
 $= 1899.212 \text{ mm}$

Determine the basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (25.4 - 19) / 25.4$
 $= 0.250$

Equivalent outer tube limit radius [ao]:

$= Do / 2$
 $= 1899/2$



$$= 949.6060 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{(1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))}$$

$$= 25.4 / \sqrt{(1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 1899 \cdot 25.4) / (\pi \cdot 1899^2))}$$

$$= 25.4000 \text{ mm}$$

Effective ligament efficiency for bending [μ*]:

$$= (p* - d*) / p*$$

$$= (25.4 - 19) / 25.4$$

$$= 0.2500$$

Ratio [ps]:

$$= as / ao$$

$$= 965/950$$

$$= 1.0162$$

Ratio [pc]:

$$= ac / ao$$

$$= 965/950$$

$$= 1.0162$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 4722 ((19 - 2 \cdot 2.1) / (2 \cdot 950))^2$$

$$= 0.7113$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 4722 (19 / (2 \cdot 950))^2$$

$$= 0.5249$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((6 - 0); 0)$$

$$= 6.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts(Ds + ts) Es / L$$

$$= \pi \cdot 28 (1930 + 28) 2067025 / 9260$$

$$= 384463.3750 \text{ kgf/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt(Dt - tt) Et / L$$

$$= \pi \cdot 2.1 (19 - 2.1) 2067025 / 9260$$

$$= 249.6164 \text{ kgf/mm}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 384463 / (4722 \cdot 250)$$

$$= 0.32618$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks / Kj)$$

$$= 1 / (1 + 384463 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [βs]:

$$= ((12(1 - \nu s^2))^{1/4}) / ((Ds + ts) ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1930 + 28) 28)$$

$$= 0.0078 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta s \cdot Es \cdot ts^3 / (6(1 - \nu s^2))$$

$$= 0.0078 \cdot 2067025 \cdot 28^3 / (6(1 - 0.3^2))$$

$$= 64520488.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Shell coefficient [λs]:



$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1930 + 28) \cdot 64520488 / 355^3 \cdot (1 + 355 \cdot 0.0078 + (355 \cdot 0.0078)^2 / 2)$$

$$= 127987.2500 \text{ kgf/cm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 1930^2 / (4 \cdot 2067025 \cdot 28) \cdot (1 - 1930 \cdot 0.3 / (2(1930 + 28)))$$

$$= 0.13711\text{E-}01 \text{ mm/kgf/cm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{ ((D_c + t_c) t_c) }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((1930 + 20) 20) }$$

$$= 0.0092 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_c^2))$$

$$= 0.0092 \cdot 2067025 \cdot 20^3 / (6(1 - 0.3^2))$$

$$= 27878318.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(1930 + 20 / \cos(0)) \cdot 27878318 / 355^3 \cdot (1 + 355 \cdot 0.0092 \cdot (1 + 1) / 2 + (355 \cdot 0.0092)^2 / 2 \cdot 1)$$

$$= 70041.0 \text{ kgf/cm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot T_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1930^2 / (4 \cdot 2067025 \cdot 20) \cdot (1 - 1930 \cdot 0.3 / (2(1930 + 20)))$$

$$= 0.19182\text{E-}01 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 13.976378 ; \mu^* = 0.250000 ; E^* = 447739 \text{ kgf/cm}^2$$

$$E^*/E = 0.216611 ; v^* = 0.396457$$

Note: As h/p (13.976) is > 2, data values for h/p = 2 were used.

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - v^{*2}) Nt \cdot Et \cdot t(dt - tt) a_o^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.4^2) 4722 \cdot 2067025 \cdot 2.1 (19 - 2.1) 950^2) / (447740 \cdot 9260 \cdot 355^3))^{1/4}$$

$$= 2.4176$$

Values from Table 4.18.3

$$Z_d = 0.098832 ; Z_v = 0.135038 ; Z_m = 0.585290$$

$$Z_a = 0.121971\text{E+}01 ; Z_w = 0.135038$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1986 / 1899$$

$$= 1.0457$$

Compute coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.4) / 447740 \cdot (127987 + 70041 + 2067025 \cdot \ln(1.05))$$

$$= 0.3914$$

Compute parameter [ϕ]:

$$= (1 + v^*) F$$

$$= (1 + 0.4) 0.39$$



= 0.5466

Compute parameter [Q1]:

$$= (p_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.02 - 1 - 0.55 \cdot 0.14) / (1 + 0.55 \cdot 0.59)$$

$$= -0.043642$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot Xa^4$$

$$= (0.099 + -0.044 \cdot 0.14) / 2 \cdot 2.42^4$$

$$= 1.5876$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot Xa^4$$

$$= (0.14 + -0.044 \cdot 0.59) / 2 \cdot 2.42^4$$

$$= 1.8704$$

Compute parameter [U]:

$$= (Z_w + (p_s - 1) Z_m) Xa^4 / (1 + \phi \cdot Z_m)$$

$$= (0.14 + (1.02 - 1) 0.59) 2.42^4 / (1 + 0.55 \cdot 0.59)$$

$$= 3.7408$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

= 0 (config a)

Compute Parameter [y]:

= 0.000 mm (For Pressure only cases)

Calculate parameter [ws]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 64520488 \cdot 0.0078 \cdot 0.013710845 (1 + 355 \cdot 0.0078)$$

$$= 26215.3535 \text{ mm}^2$$

Calculate parameter [w*s]:

$$= a o^2 (p_s^2 - 1) \cdot (p_s - 1) / 4 - w s$$

$$= 950^2 (1.02^2 - 1) \cdot (1.02 - 1) / 4 - 26215$$

$$= -26095.9062 \text{ mm}^2$$

Parameter [wc]:

$$= p_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)]$$

$$= 1.02 \cdot 27878318 [0.01918152 (1 + 0.0092 \cdot 1 \cdot 355) - 0.000 (1/355 + 1 \cdot 0.0092/2)]$$

$$= 21348.0352 \text{ mm}^2$$

Parameter [w*c]:

$$= a o^2 [(p_c^2 + 1) \cdot (p_c - 1) / 4 - (p_s - 1) / 2] - w c$$

$$= 950^2 [(1.02^2 + 1) \cdot (1.02 - 1) / 4 - (1.02 - 1) / 2] - 21348$$

$$= -21228.5879 \text{ mm}^2$$

Pressure [P*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/Kst \cdot (D_s/D_o)^2] v_s - [(p_s^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(D_j^2 - (D_s)^2)/D_o^2] \}$$

$$= 37.2 \{ 0.52 + 2(1 - 0.52) 0.3 + [2/0.33 \cdot (1930/1899)^2] 0.3 - [(1.02^2 - 1)/(1 \cdot 0.33)] - [(1 - 1)/(2 \cdot 1 \cdot 0.33)] [(0^2 - (1930)^2)/1899^2] \}$$

$$= 97.1516 \text{ kgf/cm}^2$$

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot Kst)] P_t$$



$$= [0.71 + 2(1 - 0.71)0.3 + 1/(1 + 0.33)]14.3$$
$$= 56.4362 \text{ kgf/cm}^2$$

Pressure [Py]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$
$$= 4722 \cdot 250 \cdot 0 / (3.14 \cdot 950^2)$$
$$= 0.0000 \text{ kgf/cm}^2$$

Pressure [Pw]:

$$= -\gamma b \cdot U \cdot W* / (2 \cdot \pi \cdot ao^2)$$
$$= -0 \cdot 3.74 \cdot 0 / (2 \cdot \pi \cdot 950^2)$$
$$= -0.0000 \text{ kgf/cm}^2$$

Calculate the pressure [Prim]:

$$= -(U/ao^2) \cdot (\omega*s \cdot Ps - \omega*c \cdot Pt)$$
$$= -(3.74/950^2) \cdot (-26096 \cdot 37.2 - -21229 \cdot 14.3)$$
$$= 2.7724 \text{ kgf/cm}^2$$

Calculate the pressure [Pw]:

$$= (U/ao^2) \cdot (\omega_s \cdot P*s - \omega_c \cdot P*c)$$
$$= (3.74/950^2) \cdot (26215 \cdot 0 - 21348 \cdot 0)$$
$$= 0.0000 \text{ kgf/cm}^2$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1)Qz2)) \cdot$$
$$(P's - P't + Py + Pw + Pw + Prim)$$
$$= 0.1000E+01 \cdot 0.33 / (1 + 1 \cdot 0.33(1.59 + (1.02 -$$
$$1)1.87)) \cdot (97.2 - 56.4 + 0 + 0 + -0 + 2.77)$$
$$= 9.2849 \text{ kgf/cm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega*s \cdot Ps - \omega*c \cdot Pt) - (\omega_s \cdot P*s - \omega_c \cdot P*c)) +$$
$$W* \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$
$$= [((-26096 \cdot 37.2 - -21229 \cdot 14.3) -$$
$$(26215 \cdot 0 - 21348 \cdot 0)) \cdot 0.01 +$$
$$0 \cdot 0 / (2 \cdot \pi)] / (1 + 0.55 \cdot 0.59)$$
$$= -5063.2510 \text{ kgf}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$
$$= -0.044 + 2 \cdot -5063 / (9.28 \cdot 950^2)$$
$$= -0.16459$$

Fm Value from Table 4.18.3 = 0.090555

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu*) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe$$
$$= (1.5 \cdot 0.091 / 0.25) \cdot (2 \cdot 950 / (355 - 6))^2 \cdot 9.28$$
$$= 149.3952 \text{ kgf/cm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot Sy = 1.5 \cdot 0.95 \cdot 2549 = 3633 \text{ kgf/cm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot Fm / \mu*) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe$$
$$= (1.5 \cdot 0.029 / 0.25) \cdot (2 \cdot 950 / (36.9 - 6))^2 \cdot 5.46$$
$$= 3632.5205 \text{ kgf/cm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + Cats + Catc$$
$$= 36.9 + 0 + 0$$
$$= 36.8967 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / ao) \cdot (0.8 \cdot \beta T \cdot Sy)$$
$$= (2 \cdot 0.25 \cdot 355 / 950) \cdot 1937$$



= 362.1304 kgf/cm²

Shear Stress is not required to be computed; [Pe] <= Taulimit

Could not converge on a minimum required tubesheet thk. to satisfy the Shear Stress criteria.



Use the best Estimated Required Tubesheet Thickness value as design guidance & Compare the Actual Stresses versus Allowable Stresses.

Tubesheet thickness = 7.7823 mm
Tubesheet Shear Stress = 575.8097 kgf/cm²

Required Tubesheet Thickness for Given Loadings (Including CA) [H reqd]:

= max(HreqB, HreqS)
= max(36.9, 7.78)
= 36.8967 mm

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

Ftmin = 0.56, Ftmax = 1.31

First Extreme Tube Axial Stress from among all the tubes [σ_1]:

= ((Ps • xs - Pt • xt) - Pe • Ftmin) / (Xt - Xs)
= ((37.2 • 0.52 - 14.3 • 0.71) - (9.28) • 0.56) /
(0.71 - 0.52)
= 22.3595 kgf/cm²

Second Extreme value of Tube Axial Stress from among all the tubes [σ_2]:

= ((Ps • xs - Pt • xt) - Pe • Ftmax) / (Xt - Xs)
= ((37.2 • 0.52 - 14.3 • 0.71) - (9.28) • 1.31) /
(0.71 - 0.52)
= -14.8538 kgf/cm²

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

= max(| $\sigma_{t,1}$ |; | $\sigma_{t,2}$ |) = 22.4 kgf/cm²

Allowable Tube Stress [$\sigma_{t,A}$]:

= $\beta T \cdot S_{y,t}$ = 2324.8306 kgf/cm²

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

= max((3.25 - 0.25(Zd + Q3 • Zw)Xa⁴); 1.25)
= max((3.25 - 0.25(0.099 + -0.16 • 0.14)2.42⁴); 1.25)
= 2.0000 (Should be ≤ 2)

Determine the factor [rt]:

= ($\sqrt{dt^2 + (dt - 2 \cdot tt)^2}$) / 4
= ($\sqrt{19^2 + (19 - 2 \cdot 2.1)^2}$) / 4
= 6.0385 mm

Determine the factor [Ct]:

= $\sqrt{2 \cdot \pi^2 \cdot Et / Syt}$
= $\sqrt{2 \cdot \pi^2 \cdot 2067025 / 2447}$
= 135.0872

Determine the factor [Ft]:

= k • L/r
= 1 • 470/6.04
= 77.8333

Buckling Allowable Stress [Stb]:

= (Sy,t / Fs) • (1 - Ft / (2 • Ct))



$$= (2447/2) \cdot (1 - 77.8/(2 \cdot 135))$$
$$= 2324.831 \text{ kgf/cm}^2 \text{ (Never greater than Sot)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$
$$= 22.4 \cdot 1.12$$
$$= 25.00 \text{ kgf}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$
$$= \pi (19 - 2.1) 2.1$$
$$= 1.118 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$
$$= \pi \cdot 2.1 (19 - 2.1) 2325$$
$$= 2599.738 \text{ kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$
$$= 0.85 \cdot \pi \cdot 1.3 (19 + 0.67 \cdot 1.3) 2325$$
$$= 1607.7332 \text{ kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$
$$= 2325/2325$$
$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$
$$= 0.55 \cdot \pi \cdot 1.3 (19 + 0.67 \cdot 1.3) 2325$$
$$= 1040.2980 \text{ kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$
$$= 2 \left(\sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 2.1 \cdot (19 - 2.1) 1 \cdot 1)} - 0.75 \cdot 19 \right)$$
$$= 2.552 \text{ mm}$$

$$\text{Minimum Required Fillet Weld Leg [afr]: } 1.2759 \text{ mm}$$

$$\text{Minimum Required Groove Weld Leg [agr]: } 1.2759 \text{ mm}$$

Tube-Tubesheet Jt allowable, 2600 is $\geq$ tube strength 2600 kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= \beta T \cdot S_{y,s} = 0.95 \cdot 2651 = 3778 \text{ kgf/cm}^2$$

Axial Membrane Stress in Shell [σ_s ,m]:

$$= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$
$$= 950^2 / ((1930 + 28) 28) \cdot [9.28 + (1.02^2 - 1)(37.2 - 14.3)] + 965^2 \cdot 14.3 / ((1930 + 28) 28)$$
$$= 407.7205 \text{ kgf/cm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 28 mm, the minimum Shell length adjacent to the tubesheet should be 418.437 mm



Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	149.4 <=	3632.5 kgf/cm ²	Pass
Tubesheet Shear Stress	45.6 <=	1937.4 kgf/cm ²	Pass
Maximum Tube Stress	22.4 <=	2324.8 kgf/cm ²	Pass
Minimum Tube Stress (Buckling)	-14.9 <=	-2324.8 kgf/cm ²	Pass
Maximum Force on any one Tube	25.0 <=	2599.7 kgf	Pass
Axial Membrane Stress in Shell	407.7 <=	3777.8 kgf/cm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	36.897	355.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.276	1.300	Ok
Tube-Tubesheet Groove Weld Leg :	1.276	1.300	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/ Fail
D1uc	37.412 ...	248	3633 60 1937	Ps+Pt-Th D1	Ok
D2uc	58.554 * ...	397	3633 106 1937	Ps+Pt-Th D2	Ok
D3uc	36.897 ...	149	3633 46 1937	Ps+Pt-Th D3	Ok
Max:	58.5538 ... mm	0.109	0.055 (Str. Ratio)		

The program could not converge on a required thickness value for the tubesheet. Hence, the best guess of the required thickness is provided. Check the tubesheet, tube, shell and the channel stress comparisons for acceptability.

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	----	Shell Stresses Ten Allwd Cmp Allwd	:	----	Shell Band Stress Ten Allwd Cmp Allwd	:	Pass Fail
D1uc	32	3777	...	...	...	...	Ok
D2uc	375	3777	...	...	...	...	Ok



D3uc 408 3777 Ok

Max RATIO 0.108

Tube, Shell and Channel Stress Summary:

---- Tube Stresses Tube Loads Shell Stress Channel Stress Pass											
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	44	2324	-28	-2324	49	2599	...	3777	...	3777	Ok
D2uc	59	2324	-59	-2324	66	2599	...	3777	...	3777	Ok
D3uc	22	2324	-15	-2324	25	2599	...	3777	...	3777	Ok
Max RATIO	0.025		0.025		0.025		...		...		

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	58.554	355.000	Ok
Tubesheet thickness flanged extension :	0.000	64.000	Ok
Tube thickness :	0.596	2.100	Ok
Tube-tubesheet fillet weld leg :	1.276	1.300	Ok
Tube-tubesheet groove weld leg :	1.276	1.300	Ok

Min Shell length of thk, (28.000) adj. to tubesheet: 418.437 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.198 mm
 Due to Pressure + Thermal Shell Compresses by : -0.198 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0216	...	...	...	...	...
2	-0.1982	...	...	...	...	...
3	-0.1765	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	No Calc	...	102	No Calc	No Calc
20	30	No Calc	...	107	No Calc	No Calc
30	Tsht	164.997	1970	17	0.001308	949.598
Tsht	40	No Calc	...	17	No Calc	No Calc
40	60	1076	1986	26.5	0.00033611	336.037
60	Tsht	1076	1970	17	0.00017636	179.789
Tsht	70	No Calc	...	17	No Calc	No Calc
70	80	No Calc	...	107	No Calc	No Calc
80	90	No Calc	...	102	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	105	35.3183	...	No Calc
20	30	110	86.5632	...	No Calc
30	Tsht	20	No Calc	...	10.9
Tsht	40	20	No Calc	...	No Calc
40	60	28	No Calc	...	5.98
60	Tsht	20	No Calc	...	2.07
Tsht	70	20	No Calc	...	No Calc
70	80	110	86.5632	...	No Calc
80	90	105	35.3183	...	No Calc

Minimum

2.07

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	164.997	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	60	1076	No Calc	No Calc	No Calc
60	Tsht	1076	No Calc	No Calc	No Calc
Tsht	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	3179.56	...	3179.56	...	317.956
20	30	692.33	464384	670.021	465113	69.233
30	40	1424.39	4388304	1212.6	4415631	142.439
40	60	12361	14627608	11707.8	14711891	1236.1
60	70	1424.39	2369482	1212.6	2396809	142.439
70	80	692.33	464384	670.021	465113	69.233
80	90	3179.56	...	3179.56	...	317.956
Total		22953	22314162.00	21832	22454560.00	2295



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset Dtl. Cent. cm	Description
10	Liqd	...	5.25	...	...	Liquid: 10
20	Liqd	464.175	7.5	...	...	Liquid: 20
30	Liqd	4386.33	75	...	...	Liquid: 30
30	Nozl	441.449	75	125.235	...	A1-Tube Inlet
30	Nozl	441.449	75	-125.235	...	B1-Tube Inlet
40	Sadl	540.752	185	121.15	...	Left Saddle
40	Sadl	540.752	741	121.15	...	New Saddle
40	Liqd	292.42	463	...	...	Liquid: 40
40	Nozl	124.279	80	...	108.646	A2-Shell Inlet
40	Nozl	127.459	870	...	108.646	A3-Shell Inlet
60	Liqd	2368.41	75	...	...	Liquid: 50
70	Liqd	464.175	7.5	...	...	Liquid: 60
80	Liqd	...	5.25	...	...	Liquid: 70
30	FTsh	5302.12	167.75	...	...	
30	Tube	44882.3	648.5	...	...	Tubes
30	RTsh	5302.12	1129.25	...	...	Tubesheet
40	Baf1	144.539	52	...	...	Baffles
40	Baf1	144.539	99	...	...	Baffles
40	Baf1	144.539	146	...	...	Baffles
40	Baf1	144.539	193	...	...	Baffles
40	Baf1	144.539	240	...	...	Baffles
40	Baf1	144.539	287	...	...	Baffles
40	Baf1	144.539	334	...	...	Baffles
40	Baf1	144.539	381	...	...	Baffles
40	Baf1	144.539	428	...	...	Baffles
40	Baf1	144.539	475	...	...	Baffles
40	Baf1	144.539	522	...	...	Baffles
40	Baf1	144.539	569	...	...	Baffles
40	Baf1	144.539	616	...	...	Baffles
40	Baf1	144.539	663	...	...	Baffles
40	Baf1	144.539	710	...	...	Baffles
40	Baf1	144.539	757	...	...	Baffles
40	Baf1	144.539	804	...	...	Baffles
40	Baf1	144.539	851	...	...	Baffles
40	Baf1	144.539	898	...	...	Baffles
40	Baf1	144.539	945	...	...	Baffles
40	Baf1	145.175	992	...	...	Baffles

Total Weight of Each Detail Type:

Saddles 1081.5



Liquid 7975.5
 Nozzles 1134.6
 Exchanger Components 58522.5
 Liquid in Tubes 8150.2

Sum of the Detail Weights 76864.3 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	25248.9	25248.9	25248.9	25248.9	25248.9	25248.9
Saddles	1081.5	1081.5	1081.5	1081.5	1081.5	1081.5
Nozzles	1134.6	1134.6	1134.6	1134.6	1134.6	1134.6
Exchanger	58522.5	58522.5	58522.5	58522.5	58522.5	58522.5
Ope. Liquid	...	...	...	...	...	7975.5
Tube Ope Lqd	...	...	...	...	...	8150.2
Test Liquid	...	22304.1	...	...	...	...
Tube Tst Lqd	...	8150.2	...	...	...	...
Totals	85987.5	116441.9	85987.5	85987.5	85987.5	102113.2

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt. - Bare weight without removable internals 85987.5 kg
 Shop Test Wt. - Fabricated weight + water (full) 116441.9 kg
 Shipping Wt. - Fab. weight + removable intls.+ shipping app. 85987.5 kg
 Erected Wt. - Fab. wt + or - loose items (trays,platforms etc.) 85987.5 kg
 Ope. Wt. no Liq - Fab. weight + internals. + details + weights 85987.5 kg
 Operating Wt. - Empty weight + operating liq. uncorroded 102113.2 kg
 Oper. Wt. + CA - Corr wt. + operating liquid 100879.7 kg
 Field Test Wt. - Empty weight + test medium (full) 116441.9 kg

Exchanger Tube Data

Volume of Exchanger tubes : 8153892.0 cm³
 Weight of Ope Liq in tubes : 8150.2 kg
 Weight of Water in tubes : 8150.2 kg

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	43937.5	...
20	30	17520.7	9579.97
30	40	92834.1	90949.1
40	60	577750	561459
60	70	92834.1	90949.1
70	80	17520.7	9579.97
80	90	43937.5	...

Total 886334.9 762517.4 cm²
 Total 954.0 820.8 ft²

Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
A1-Tube Inlet	48.57	52.01	85	300	GR 1.1	1.097	48.571	48.571	48.571
B1-Tube Inlet	48.57	52.01	85	300	GR 1.1	1.097	48.571	48.571	48.571
A2-Shell Inlet	45.20	52.01	180	300	GR 1.1	2.154	45.195	45.195	45.195
A3-Shell Inlet	45.20	52.01	180	300	GR 1.1	2.156	45.195	45.195	45.195

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 45.195 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 52.007 kgf/cm²

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 48.571 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 52.007 kgf/cm²

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME kgf-m.	Axial Force FE kgf	MAWP PD kgf/cm ²	Maximum Pressure Pmax kgf/cm ²	Pass Fail
A1-Tube Inlet	657.225	0.50	8	3671	10.868	48.571	Pass
B1-Tube Inlet	657.225	0.50	8	3671	10.868	48.571	Pass
A2-Shell Inlet	302.387	0.50	1	1530	24.261	45.195	Pass
A3-Shell Inlet	302.387	0.50	1	1530	24.261	45.195	Pass

Evaluating Flange on Nozzle [A1-Tube Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(8) + 4 \cdot \text{abs}(3671)657.225 \leq \pi \cdot 657.225^3 \cdot ((48.571 - 10.868) + .500 \cdot 48.571)$$

$$= 0.97807E+04 \leq 0.55285E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle A1-Tube Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(48.571, 0.500 \cdot 48.571 + 48.571 - 9780.7 / (\pi \cdot 657.225^3))$$

$$= \min(48.571, 71.760)$$

$$= 48.571 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [B1-Tube Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(8) + 4 \cdot \text{abs}(3671)657.225 \leq \pi \cdot 657.225^3 \cdot ((48.571 - 10.868) + .500 \cdot 48.571)$$

$$= 0.97807E+04 \leq 0.55285E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle B1-Tube Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(48.571, 0.500 \cdot 48.571 + 48.571 - 9780.7 / (\pi \cdot 657.225^3))$$

$$= \min(48.571, 71.760)$$

$$= 48.571 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [A2-Shell Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1) + 4 \cdot \text{abs}(1530)302.387 \leq \pi \cdot 302.387^3 \cdot ((45.195 - 24.261) + .500 \cdot 45.195)$$

$$= 0.18707E+04 \leq 0.37813E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle A2-Shell Inlet [Pmax]:



$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(45.195, 0.500 \cdot 45.195 + 45.195 - 1870.7 / (\pi \cdot 302.387^3)) \\ &= \min(45.195, 65.639) \\ &= 45.195 \text{ kgf/cm}^2 \end{aligned}$$

Evaluating Flange on Nozzle [A3-Shell Inlet]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(1) + 4 \cdot \text{abs}(1530) 302.387 \leq \pi \cdot 302.387^3 [(45.195 - 24.261) + 0.500 \cdot 45.195] \\ &= 0.18727\text{E}+04 \leq 0.37813\text{E}+05 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle A3-Shell Inlet [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(45.195, 0.500 \cdot 45.195 + 45.195 - 1872.7 / (\pi \cdot 302.387^3)) \\ &= \min(45.195, 65.637) \\ &= 45.195 \text{ kgf/cm}^2 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	0 cm
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 cm
Distance Upwind of Crest Lh	0 cm
Distance from Crest to the Vessel x	0 cm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 cm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.19 \cdot 0.98) / (1 + 1.7 \cdot 3.4 \cdot 0.19)))$
 $= \min(0.85, 0.92)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.601
Structure Height to Diameter ratio	7.180

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
Exposure Category: D from Table 26.11-1
 $\alpha = 11.5 : Zg = 58979 \text{ cm}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 150 + 0 = 150 \text{ cm}$
 $= 4.92 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[Kz]$:
Because z (4.92 ft.) $< 15 \text{ ft.}$
 $= 2.41(15 / Zg)^{2/\alpha}$
 $= 2.41(15/1935)^{2/11.5}$
 $= 1.04$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:
 $K1 = 0, K2 = 0, K3 = 0$



Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.04 \cdot 1 \cdot 1 \cdot 1 \cdot 112^2)$$

$$= 33.1 \text{ psf } [162 \text{ kgf/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.1 \cdot 0.85 \cdot 0.6 \cdot 2.92$$

$$= 49.4 \text{ lbs. } [22.4 \text{ kgf}]$$

Element	Hgt (z) cm	K1	K2	K3	Kz	Kzt	qz kgf/m ²
Lhs Blind flang	150.0	0.000	0.000	0.000	1.035	1.000	161.847
LHS Ch Flg	150.0	0.000	0.000	0.000	1.035	1.000	161.847
LHS Ch Shell	150.0	0.000	0.000	0.000	1.035	1.000	161.847
Main shell-1	150.0	0.000	0.000	0.000	1.035	1.000	161.847
RHS Ch shell	150.0	0.000	0.000	0.000	1.035	1.000	161.847
RHS Ch Flg	150.0	0.000	0.000	0.000	1.035	1.000	161.847
RHS Blind Flg	150.0	0.000	0.000	0.000	1.035	1.000	161.847

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	150	258	2709	161.847	24.6377
20	30	150	231.6	3474	161.847	31.5952
30	40	150	236.4	35460	161.847	322.5
40	60	150	238.32	220684	161.847	2007.07
60	70	150	236.4	35460	161.847	322.5
70	80	150	231.6	3474	161.847	31.5952
80	90	150	258	2709	161.847	24.6377

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio	z/h
Amplification Factor	Ap
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR]
Component strength factor	[Rpo]
Overstrength factor	[Omega0]
Response Acceleration Factor	[SMS]
Response Acceleration Factor	[SM1]
Building Importance Factor	[Ie]
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.33 = 0.22$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.14 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(8.450.75)$
 = 0.099 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.099$, 0.03) per 12.8.2
 = 0.030

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.22 \cdot 99798 \cdot 1.5$
 = 9760.263 kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.22 \cdot 99798$
 = 4337.9 kgf

Earthquake Load Calculation:

From	To	Earthquake Height cm	Earthquake Weight kgf	Element Ope Load kgf
10	20	107.5	11088.7	1178.25
20	30	96.5	11088.7	1057.68
30	40	96.5	11088.7	1057.68
40	Sad1	96.5	11088.7	1057.68
Sad1	60	96.5	11088.7	1057.68
40	60	96.5	11088.7	1057.68



60	70	96.5	11088.7	1057.68
70	80	96.5	11088.7	1057.68
80	90	107.5	11088.7	1178.25

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	674.318 cm
Center of Gravity of the Liquid	518.691 cm
Center of Gravity of the Nozzles	231.828 cm
Center of Gravity of the Tubesheet(s)	674.318 cm
Center of Gravity of the Tubes	674.317 cm
Center of Gravity of the Baffles	733.416 cm
Center of Gravity of Bare Shell New and Cold	672.168 cm
Center of Gravity of Bare Shell Corroded	672.402 cm
Vessel CG in the Operating Condition	658.359 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	669.934 cm
Vessel CG in the Test Condition	660.421 cm

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	979.75	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	1076.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1850.00	mm
Saddle Width	b1 or b	275.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	550.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1500.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		67554.28	kgf
Pressure used in Analysis	P	25.495	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	396.53	1407.13
Long. Stress at Bottom of Midspan	546.08	1407.13
Long. Stress at Top of Saddles	688.14	1407.13
Long. Stress at Bottom of Saddles	351.12	1407.13
Tangential Shear in Shell	199.86	1125.71
Circ. Stress at Horn of Saddle	405.73	1758.92
Circ. Compressive Stress in Shell	24.09	1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (2765/2 + 0) \cdot 1500/1744$
 $= 3566.3 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(794, 0) \cdot 1500/5560$
 $= 214.3 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(9760, 0) \cdot 1500/5560$
 $= 2633.2 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (9760/2 + 0) \cdot 1500/1744$
 $= 12590.8 \text{ kgf}$



Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 54963 + \max(214, 3566, 2633, 12591)$$

$$= 67554.3 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587 (0 + \pi/4(1.99 \cdot 1.2)^2)$$

$$= 7788.071 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	68095.03	kgf
Transverse Shear Load	4880.13	kgf
Longitudinal Shear Load	9760.26	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(1930/2 + 28 + 14)\sin(120/2), 2 \cdot 980)$$

$$= 1744.175 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(\text{L} \cdot \text{a})) / (1 + (4\text{hm})/3\text{L})]$$

$$= -67554 \cdot 185 [1 - (1 - 185/1076 + (98^2 - 0^2)/(2 \cdot 185 \cdot 1076)) / (1 + (4 \cdot 0)/(3 \cdot 1076))]$$

$$= -18474.1 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm}^2)/3\text{L}) - 4\text{a}/\text{L})$$

$$= 67554 \cdot 1076/4((1 + 2(98^2 - 0^2)/(1076^2)) / (1 + (4 \cdot 0)/(3 \cdot 1076)) - 4 \cdot 185/1076)$$

$$= 59759.0 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot \text{Rm}/(2t) - M2/(\pi \cdot \text{Rm}^2 t)$$

$$= 25.5 \cdot 980/(2 \cdot 26.5) - 59759/(\pi \cdot 980^2 \cdot 26.5)$$

$$= 396.53 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot \text{Rm}/(2t) + M2/(\pi \cdot \text{Rm}^2 \cdot t)$$

$$= 25.5 \cdot 980/(2 \cdot 26.5) + 59759/(\pi \cdot 980^2 \cdot 26.5)$$

$$= 546.08 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot \text{Rm}/(2t) - M1/(K1 \cdot \pi \cdot \text{Rm}^2 t)$$

$$= 25.5 \cdot 980/(2 \cdot 26.5) - 18474/(0.11 \cdot \pi \cdot 980^2 \cdot 26.5)$$

$$= 688.14 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot \text{Rm}/(2t) + M1/(K1^* \cdot \pi \cdot \text{Rm}^2 \cdot t)$$

$$= 25.5 \cdot 980/(2 \cdot 26.5) + 18474/(0.19 \cdot \pi \cdot 980^2 \cdot 26.5)$$

$$= 351.12 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(\text{L} - 2\text{a})/(\text{L} + (4 \cdot \text{hm}/3))$$

$$= 67554 (1076 - 2 \cdot 185)/(1076 + (4 \cdot 0/3))$$

$$= 44324.7 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:



$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 44325 / (980 \cdot 26.5)$$

$$= 199.86 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{ (Rm \cdot t) }$$

$$= 0.78 \cdot \sqrt{ (980 \cdot 26.5) }$$

$$= 125.683 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{ (Rm \cdot t) }, 2a)$$

$$= \min(275 + 1.56 \cdot \sqrt{ (980 \cdot 26.5) }, 2 \cdot 185)$$

$$= 526.37 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(1407/1407, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 67554 \cdot 0.1 / (526 (26.5 + 1 \cdot 14))$$

$$= -24.09 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr) b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -67554 / (4(26.5 + 1 \cdot 14) 526) -$$

$$3 \cdot 0.053 \cdot 67554 / (2(26.5 + 1 \cdot 14)^2)$$

$$= -405.73 \text{ kgf/cm}^2$$

Distance between Saddle Supports [Ls]:

$$= 5560.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot Ls(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.12620\text{E-}04 \cdot 5560 (180 - 21.1)$$

$$= 11.149 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1800.0000	mm
Baseplate Thickness	Bpthk	24.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2318.7	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{ (R \cdot t) } + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{ (966 \cdot 26.5) } + 550) 26.5^3) / 12$$

$$= 124.011 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 124 + 212 (143 - 13.3)^2$$

$$= 35789.227 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	799.7	26.5	13.3	211.9	280780.5	0.358E+05
Wearplate	550.0	14.0	33.5	77.0	257950.0	0.924E+04
Web	14.0	469.0	275.0	65.7	1805649.8	0.235E+05
BasePlate	300.0	24.0	521.5	72.0	3754799.5	0.103E+06

Totals	...	...	...	426.6	6099180.0	0.172E+06
--------	-----	-----	-----	-------	-----------	-----------

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6099180/427$$

$$= 142.982 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 67554$$

$$= 13748.7607 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 64.0491 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1391.2390 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1500 - 966 \cdot \sin(120/2) / 1.05$$

$$= 700.711 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 9633.9170 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 80.2241 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1545.8213 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(67554 + 541)300 / (4 \cdot 1800 \cdot 1546)}$$

$$= 23.466 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (965 + 28 + 14)$$

$$= 1744.175 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1744 / (4 - 1)$$

$$= 581.392 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 581 \cdot 300 / 2$$

$$= 872.088 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 68095 / (1800 \cdot 300)$$

$$= 12.610 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 872 \cdot 12.6$$

$$= 10997.191 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 36.5 + 40.7$$



$$= 77.237 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 10997/77.2$$

$$= 142.382 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	275.0	...	38.5	...	0.243E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 275 - 14$$

$$= 261.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/77.2$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 275/2$$

$$= 137.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(2426/77.2)}$$

$$= 56.048 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1500 - \cos(120/2) (965 + 28 + 14) - 24$$

$$= 972.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038900/2319)}$$

$$= 131.746$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 972/56$$

$$= 17.351$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 9760 / (2 \cdot 1800) \cdot 581 \cdot 972/2$$

$$= 766.457 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($17.4 < 132$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (17.4)^2 / (2 \cdot 132^2)) 2319 /$$

$$(5/3 + 3 \cdot (17.4) / (8 \cdot 132) - (17.4^3) / (8 \cdot 132^3))$$

$$= 1340 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 142/1340 + (766 \cdot 138/24263022) / 1546$$

$$= 0.387$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
--	---	---	---	---	----	----



Rib	14.0	261.0	0.0	36.5	0.0	0.243E+04
Web	581.4	14.0	0.0	81.4	0.0	13.3
Totals	...	...	...	117.9	...	0.244E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/118$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 275/2$$

$$= 137.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1500 - \sqrt{(1007 + 14)^2 - (581/2)^2} - 24$$

$$= 511.871 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(2439/118)}$$

$$= 45.479 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 512/45.5$$

$$= 11.255$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 9760 / (2 \cdot 1800)$$

$$= 2.711 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 2.71 \cdot 581 \cdot 512$$

$$= 806.843 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 807/177402$$

$$= 454.811 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($11.3 < 132$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.3)^2 / (2 \cdot 132^2)) 2319 /$$

$$(5/3 + 3 \cdot (11.3) / (8 \cdot 132) - (11.3^3) / (8 \cdot 132^3))$$

$$= 1360 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 187/1360 + (807 \cdot 138/2439) / 1546$$

$$= 0.431$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1529.00 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	3.0418 cm ²
Saddle Load Q0 (Weight)	Q0	55504.2 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	2633.2 kgf



Maximum Transverse Force	Ft	4880.1 kgf
Maximum Longitudinal Force	Fl	9760.3 kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
= 3977.1 kgf

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
= max(Ff, Fl) / (Bolt Area • Number of Bolts)
= 9760 / (3.04 • 4)
= 802.2 kgf/cm². Must be less than 1019.0 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
= Ft / (Bolt Area • Number of Bolts)
= 4880 / (3.04 • 4)
= 401.1 kgf/cm². Must be less than 1019.0 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= Fl / (BoltShearAllowable • Nbolts)
= 9760 / (1019 • 4)
= 2.3946 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= B • Ft + Sum of X Moments
= 1500 • 4880 + 0
= 7320.21 kgf-m.

Eccentricity (e):
= Rmom / QO
= 7320/55504
= 131.89 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0, 2.39, 0]
= 2.3946 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 14, c = 1.5 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.64, Temp. Reduction = 20 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	979.75	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	1076.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1850.00	mm
Saddle Width	b1 or b	275.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees



Wear Plate Width	b2 or b1	550.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1500.00	mm
Coefficient of Friction	mu	0.08	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		61763.42	kgf
Pressure used in Analysis	P	25.495	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	402.94	1407.13
Long. Stress at Bottom of Midspan	539.67	1407.13
Long. Stress at Top of Saddles	669.55	1407.13
Long. Stress at Bottom of Saddles	361.42	1407.13
Tangential Shear in Shell	182.73	1125.71
Circ. Stress at Horn of Saddle	370.95	1758.92
Circ. Compressive Stress in Shell	22.03	1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (2765/2 + 0) \cdot 1500/1744$
 $= 3566.3 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(794, 0) \cdot 1500/5560$
 $= 214.3 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(9760, 0) \cdot 1500/5560$
 $= 2633.2 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (9760/2 + 0) \cdot 1500/1744$
 $= 12590.8 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 49173 + \max(214, 3566, 2633, 12591)$
 $= 61763.4 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587 (0 + \pi/4 (1.99 \cdot 1.2)^2)$
 $= 7788.071 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	62304.17	kgf
Transverse Shear Load	4880.13	kgf
Longitudinal Shear Load	9760.26	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1



Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1930/2 + 28 + 14)\sin(120/2), 2 \cdot 980)$$

$$= 1744.175 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -61763 \cdot 185 [1 - (1 - 185/1076 + (98^2 - 0^2)/(2 \cdot 185 \cdot 1076))/(1 + (4 \cdot 0)/(3 \cdot 1076))]$$

$$= -16890.5 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L]$$

$$= 61763 \cdot 1076/4 [(1 + 2(98^2 - 0^2)/(1076^2))/(1 + (4 \cdot 0)/(3 \cdot 1076)) - 4 \cdot 185/1076]$$

$$= 54636.3 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 25.5 \cdot 980/(2 \cdot 26.5) - 54636/(\pi \cdot 980^2 \cdot 26.5)$$

$$= 402.94 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 25.5 \cdot 980/(2 \cdot 26.5) + 54636/(\pi \cdot 980^2 \cdot 26.5)$$

$$= 539.67 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 25.5 \cdot 980/(2 \cdot 26.5) - 16890/(0.11 \cdot \pi \cdot 980^2 \cdot 26.5)$$

$$= 669.55 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 25.5 \cdot 980/(2 \cdot 26.5) + 16890/(0.19 \cdot \pi \cdot 980^2 \cdot 26.5)$$

$$= 361.42 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 61763 (1076 - 2 \cdot 185)/(1076 + (4 \cdot 0/3))$$

$$= 40525.1 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 40525 / (980 \cdot 26.5)$$

$$= 182.73 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(980 \cdot 26.5)}$$

$$= 125.683 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(275 + 1.56 \cdot \sqrt{(980 \cdot 26.5)}, 2 \cdot 185)$$

$$= 526.37 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r/S, 1)$$

$$= \min(1407/1407, 1)$$



= 1.0000

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) $[\sigma_6, r]$:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 61763 \cdot 0.1 / (526(26.5 + 1 \cdot 14))$$

$$= -22.03 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) $[\sigma_7, r]$:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -61763 / (4(26.5 + 1 \cdot 14)526) -$$

$$3 \cdot 0.053 \cdot 61763 / (2(26.5 + 1 \cdot 14)^2)$$

$$= -370.95 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1800.0000	mm
Baseplate Thickness	Bpthk	24.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2318.7	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.080	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(966 \cdot 26.5)} + 550)26.5^3) / 12$$

$$= 124.011 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 124 + 212(143 - 13.3)^2$$

$$= 35789.227 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	799.7	26.5	13.3	211.9	280780.5	0.358E+05
Wearplate	550.0	14.0	33.5	77.0	257950.0	0.924E+04
Web	14.0	469.0	275.0	65.7	1805649.8	0.235E+05
BasePlate	300.0	24.0	521.5	72.0	3754799.5	0.103E+06
Totals	...	...	...	426.6	6099180.0	0.172E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6099180 / 427$$

$$= 142.982 \text{ mm}$$

Angle $[\beta]$:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 61763$$

$$= 12570.1953 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 58.5587 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1391.2390 \text{ kgf/cm}^2$$



Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 1500 - 966 \cdot \sin(120/2) / 1.05$
 $= 700.711 \text{ mm}$

Bending Moment, $M = F_h \cdot d = 8808.0820 \text{ kgf-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 73.3471 \text{ kgf/cm}^2$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 1545.8213 \text{ kgf/cm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$
 $= \sqrt{3(61763 + 541) \cdot 300 / (4 \cdot 1800 \cdot 1546)}$
 $= 22.446 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (965 + 28 + 14)$
 $= 1744.175 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nribs} - 1)$
 $= 1744 / (4 - 1)$
 $= 581.392 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 581 \cdot 300 / 2$
 $= 872.088 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 62304 / (1800 \cdot 300)$
 $= 11.538 \text{ kgf/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 872 \cdot 11.5$
 $= 10061.980 \text{ kgf}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 36.5 + 40.7$
 $= 77.237 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 10062 / 77.2$
 $= 130.274 \text{ kgf/cm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	275.0	...	38.5	...	0.243E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 275 - 14$
 $= 261.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 77.2$
 $= 0.000 \text{ mm}$



Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 275/2$$

$$= 137.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(2426/77.2)}$$

$$= 56.048 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1500 - \cos(120/2) (965 + 28 + 14) - 24$$

$$= 972.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038900/2319)}$$

$$= 131.746$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 972/56$$

$$= 17.351$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 9760 / (2 \cdot 1800) \cdot 581 \cdot 972/2$$

$$= 766.457 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < Cc$ ($17.4 < 132$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (17.4)^2 / (2 \cdot 132^2)) 2319 /$$

$$(5/3 + 3 \cdot (17.4) / (8 \cdot 132) - (17.4)^3 / (8 \cdot 132^3))$$

$$= 1340 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 130/1340 + (766 \cdot 138/24263022) / 1546$$

$$= 0.378$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	261.0	0.0	36.5	0.0	0.243E+04
Web	581.4	14.0	0.0	81.4	0.0	13.3
Totals	...	...	...	117.9	...	0.244E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/118$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 275/2$$

$$= 137.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1500 - \sqrt{(1007 + 14)^2 - (581/2)^2} - 24$$

$$= 511.871 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(2439/118)}$$

$$= 45.479 \text{ mm}$$



Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 512/45.5$$

$$= 11.255$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 9760 / (2 \cdot 1800)$$

$$= 2.711 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 2.71 \cdot 581 \cdot 512$$

$$= 806.843 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 807/177402$$

$$= 454.811 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($11.3 < 132$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.3)^2 / (2 \cdot 132^2)) 2319 /$$

$$(5/3 + 3 \cdot (11.3) / (8 \cdot 132) - (11.3^3) / (8 \cdot 132^3))$$

$$= 1360 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 171/1360 + (807 \cdot 138/2439) / 1546$$

$$= 0.420$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1529.00 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	3.0418 cm ²
Saddle Load QO (Weight)	QO	49713.4 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	2633.2 kgf
Maximum Transverse Force	Ft	4880.1 kgf
Maximum Longitudinal Force	Fl	9760.3 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 9760 / (3.04 \cdot 4)$$

$$= 802.2 \text{ kgf/cm}^2. \text{ Must be less than } 1019.0 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 4880 / (3.04 \cdot 4)$$

$$= 401.1 \text{ kgf/cm}^2. \text{ Must be less than } 1019.0 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= FI / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 9760 / (1019 \cdot 4)$$

$$= 2.3946 \text{ cm}^2$$



Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1500 \cdot 4880 + 0$$

$$= 7320.21 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 7320/49713$$

$$= 147.25 \text{ mm} < Bplen/6 \rightarrow \text{No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 2.39, 0]$$

$$= 2.3946 \text{ cm}^2$$

MDMT Calculations for: New Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 14, c = 1.5 mm, E* = 1

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.64$, Temp. Reduction = 20 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -35 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	979.00	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	1076.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1850.00	mm
Saddle Width	b1 or b	275.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	550.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		2386.01	kgf/cm ²
Head Allowable Stress used in Calculation		2386.01	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1500.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		61767.34	kgf
Pressure used in Analysis	P	37.330	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	587.80	2386.01
Long. Stress at Bottom of Midspan	717.41	2386.01
Long. Stress at Top of Saddles	840.58	2386.01
Long. Stress at Bottom of Saddles	548.42	2386.01
Tangential Shear in Shell	173.08	1413.05
Circ. Stress at Horn of Saddle	346.54	3579.01
Circ. Compressive Stress in Shell	20.97	2386.01

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (912/2 + 0) \cdot 1500/1744$
 $= 1176.9 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(262, 0) \cdot 1500/5560$
 $= 70.7 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 60590 + \max(70.7, 1177, 0, 0)$
 $= 61767.3 \text{ kgf}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587 (0 + \pi/4 (1.99 \cdot 1.2)^2)$
 $= 7788.071 \text{ N}$



Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	62308.09	kgf
Transverse Shear Load	456.15	kgf
Longitudinal Shear Load	262.07	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1930/2 + 28 + 14)\sin(120/2), 2 \cdot 979)$$

$$= 1744.175 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -61767 \cdot 185 [1 - (1 - 185/1076 + (97.9^2 - 0^2)/(2 \cdot 185 \cdot 1076))/(1 + (4 \cdot 0)/(3 \cdot 1076))]$$

$$= -16895.8 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L]$$

$$= 61767 \cdot 1076/4 [(1 + 2(97.9^2 - 0^2)/(1076^2))/(1 + (4 \cdot 0)/(3 \cdot 1076)) - 4 \cdot 185/1076]$$

$$= 54635.6 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 37.3 \cdot 979/(2 \cdot 28) - 54636/(\pi \cdot 979^2 \cdot 28)$$

$$= 587.80 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 37.3 \cdot 979/(2 \cdot 28) + 54636/(\pi \cdot 979^2 \cdot 28)$$

$$= 717.41 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 37.3 \cdot 979/(2 \cdot 28) - 16896/(0.11 \cdot \pi \cdot 979^2 \cdot 28)$$

$$= 840.58 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M_1/(K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 37.3 \cdot 979/(2 \cdot 28) + 16896/(0.19 \cdot \pi \cdot 979^2 \cdot 28)$$

$$= 548.42 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 61767 (1076 - 2 \cdot 185)/(1076 + (4 \cdot 0/3))$$

$$= 40527.6 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 40528 / (979 \cdot 28)$$

$$= 173.08 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(979 \cdot 28)}$$

$$= 129.141 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:



$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(275 + 1.56 \cdot \sqrt{(979 \cdot 28)}, 2 \cdot 185)$$

$$= 533.28 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:
 = 1.0000 Materials are the same, test case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 61767 \cdot 0.1 / (533 (28 + 1 \cdot 14))$$

$$= -20.97 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr) b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -61767 / (4(28 + 1 \cdot 14) 533) -$$

$$3 \cdot 0.053 \cdot 61767 / (2(28 + 1 \cdot 14)^2)$$

$$= -346.54 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1800.0000	mm
Baseplate Thickness	Bpthk	24.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(965 \cdot 28)} + 550) 28^3) / 12$$

$$= 147.523 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 148 + 226 (140 - 14)^2$$

$$= 35999.102 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	806.4	28.0	14.0	225.8	316120.2	0.360E+05
Wearplate	550.0	14.0	35.0	77.0	269500.0	0.850E+04
Web	14.0	469.0	276.5	65.7	1815498.8	0.243E+05
BasePlate	300.0	24.0	523.0	72.0	3765599.5	0.106E+06
Totals	...	...	...	440.5	6166718.5	0.174E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6166718 / 440$$

$$= 140.006 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 61767$$

$$= 12570.9922 \text{ kgf}$$



Tension Stress, $St = (Fh/As) = 58.5624 \text{ kgf/cm}^2$
 Allowed Stress, $Sa = 0.6 * \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 1500 - 965 \cdot \sin(120/2) / 1.05$
 $= 701.951 \text{ mm}$

Bending Moment, $M = Fh * d = 8824.2354 \text{ kgf-m.}$

Bending Stress, $Sb = (M * C1 / I) = 70.8329 \text{ kgf/cm}^2$
 Allowed Stress, $Sa = 2/3 * \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{(3(61767 + 541)300 / (4 * 1800 * 1767))}$
 $= 20.992 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 120/2) (965 + 28 + 14)$
 $= 1744.175 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1744 / (4 - 1)$
 $= 581.392 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 581 * 300 / 2$
 $= 872.088 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 62308 / (1800 * 300)$
 $= 11.539 \text{ kgf/cm}^2$

Axial Load [P]:
 $= Ap * Bp$
 $= 872 * 11.5$
 $= 10062.613 \text{ kgf}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 36.5 + 40.7$
 $= 77.237 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P/Ar$
 $= 10063 / 77.2$
 $= 130.282 \text{ kgf/cm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	275.0	...	38.5	...	0.243E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 275 - 14$
 $= 261.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:



$$= AY / A$$

$$= 0/77.2$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 275/2$$

$$= 137.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(2426/77.2)}$$

$$= 56.048 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1500 - \cos(120/2) (965 + 28 + 14) - 24$$

$$= 972.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038900/2651)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 972/56$$

$$= 17.351$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 262 / (2 \cdot 1800) \cdot 581 \cdot 972/2$$

$$= 20.580 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < Cc$ (17.4 < 123) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (17.4)^2 / (2 \cdot 123^2)) 2651 / (5/3 + 3 \cdot (17.4) / (8 \cdot 123) - (17.4)^3 / (8 \cdot 123^3))$$

$$= 1527 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 130/1527 + (20.6 \cdot 138/24263022) / 1767$$

$$= 0.092$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	261.0	0.0	36.5	0.0	0.243E+04
Web	581.4	14.0	0.0	81.4	0.0	13.3
Totals	...	...	...	117.9	...	0.244E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/118$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 275/2$$

$$= 137.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1500 - \sqrt{(1007 + 14)^2 - (581/2)^2} - 24$$

$$= 511.871 \text{ mm}$$

Radius of Gyration [r]:



$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{2439/118}$$

$$= 45.479 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 512/45.5$$

$$= 11.255$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 262 / (2 \cdot 1800)$$

$$= 0.073 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.073 \cdot 581 \cdot 512$$

$$= 21.665 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 21.7/177402$$

$$= 12.212 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($11.3 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.3)^2 / (2 \cdot 123^2)) 2651 /$$

$$(5/3 + 3 \cdot (11.3) / (8 \cdot 123) - (11.3^3) / (8 \cdot 123^3))$$

$$= 1552 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 171/1552 + (21.7 \cdot 138/2439) / 1767$$

$$= 0.117$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	3.0418 cm ²
Saddle Load Q0 (Weight)	Q0	61131.2 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	70.7 kgf
Maximum Transverse Force	Ft	456.1 kgf
Maximum Longitudinal Force	Fl	262.1 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 262 / (3.04 \cdot 4)$$

$$= 21.5 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 456 / (3.04 \cdot 4)$$

$$= 37.5 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= FI / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 262 / (1052 \cdot 4)$$



= 0.0623 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= B • Ft + Sum of X Moments

= 1500 • 456 + 0

= 684.22 kgf-m.

Eccentricity (e):

= Rmom / QO

= 684/61131

= 11.19 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0, 0.062, 0]

= 0.0623 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	979.00	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	1076.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1850.00	mm
Saddle Width	b1 or b	275.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	550.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		2386.01	kgf/cm ²
Head Allowable Stress used in Calculation		2386.01	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1500.00	mm
Coefficient of Friction	mu	0.08	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		55946.75	kgf
Pressure used in Analysis	P	37.330	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		593.91	2386.01
Long. Stress at Bottom of Midspan		711.30	2386.01
Long. Stress at Top of Saddles		822.87	2386.01
Long. Stress at Bottom of Saddles		558.23	2386.01
Tangential Shear in Shell		156.77	1413.05
Circ. Stress at Horn of Saddle		313.88	3579.01
Circ. Compressive Stress in Shell		18.99	2386.01

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E



$$= 3 (912/2 + 0) \cdot 1500/1744$$

$$= 1176.9 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(262, 0) \cdot 1500/5560$$

$$= 70.7 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 54770 + \max(70.7, 1177, 0, 0)$$

$$= 55946.8 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587 (0 + \pi/4 (1.99 \cdot 1.2)^2)$$

$$= 7788.071 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	56487.51	kgf
Transverse Shear Load	456.15	kgf
Longitudinal Shear Load	262.07	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1930/2 + 28 + 14) \sin(120/2), 2 \cdot 979)$$

$$= 1744.175 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h m^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -55947 \cdot 185 [1 - (1 - 185/1076 + (97.9^2 - 0^2) / (2 \cdot 185 \cdot 1076)) / (1 + (4 \cdot 0) / (3 \cdot 1076))]$$

$$= -15303.6 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(R_m^2 - h m^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 55947 \cdot 1076 / 4 ((1 + 2(97.9^2 - 0^2)/(1076^2)) / (1 + (4 \cdot 0) / (3 \cdot 1076)) - 4 \cdot 185/1076)$$

$$= 49487.1 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 37.3 \cdot 979 / (2 \cdot 28) - 49487 / (\pi \cdot 979^2 \cdot 28)$$

$$= 593.91 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 37.3 \cdot 979 / (2 \cdot 28) + 49487 / (\pi \cdot 979^2 \cdot 28)$$

$$= 711.30 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 37.3 \cdot 979 / (2 \cdot 28) - 15304 / (0.11 \cdot \pi \cdot 979^2 \cdot 28)$$

$$= 822.87 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 37.3 \cdot 979 / (2 \cdot 28) + 15304 / (0.19 \cdot \pi \cdot 979^2 \cdot 28)$$



$$= 558.23 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 55947 (1076 - 2 \cdot 185)/(1076 + (4 \cdot 0/3))$$

$$= 36708.6 \text{ kgf}$$

Shear Stress in the shell no ribs, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 36709 / (979 \cdot 28)$$

$$= 156.77 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(979 \cdot 28)}$$

$$= 129.141 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(275 + 1.56 \cdot \sqrt{(979 \cdot 28)}, 2 \cdot 185)$$

$$= 533.28 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 55947 \cdot 0.1 / (533(28 + 1 \cdot 14))$$

$$= -18.99 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q/(4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q/(2(t + \eta \cdot tr)^2)$$

$$= -55947/(4(28 + 1 \cdot 14)533) -$$

$$3 \cdot 0.053 \cdot 55947/(2(28 + 1 \cdot 14)^2)$$

$$= -313.88 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1800.0000	mm
Baseplate Thickness	Bpthk	24.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.080	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(965 \cdot 28)} + 550)28^3)/12$$

$$= 147.523 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 148 + 226(140 - 14)^2$$

$$= 35999.102 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	806.4	28.0	14.0	225.8	316120.2	0.360E+05
Wearplate	550.0	14.0	35.0	77.0	269500.0	0.850E+04
Web	14.0	469.0	276.5	65.7	1815498.8	0.243E+05
BasePlate	300.0	24.0	523.0	72.0	3765599.5	0.106E+06
Totals	...	...	...	440.5	6166718.5	0.174E+06



Distance to Centroid [C1]:

$$\begin{aligned} &= AY / A \\ &= 6166718/440 \\ &= 140.006 \text{ mm} \end{aligned}$$

Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 120/2 \\ &= 120.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.2 \cdot 55947 \\ &= 11386.3770 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh/As) = 53.0439 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians}) \\ &= 1500 - 965 \cdot \sin(120/2) / 1.05 \\ &= 701.951 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 7992.6924 \text{ kgf-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 64.1580 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{ (3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})) } \\ &= \sqrt{ (3(55947 + 541) 300 / (4 \cdot 1800 \cdot 1767)) } \\ &= 19.988 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120/2) (965 + 28 + 14) \\ &= 1744.175 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 1744 / (4 - 1) \\ &= 581.392 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 581 \cdot 300 / 2 \\ &= 872.088 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 56488 / (1800 \cdot 300) \\ &= 10.461 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 872 \cdot 10.5 \\ &= 9122.604 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 36.5 + 40.7 \\ &= 77.237 \text{ cm}^2 \end{aligned}$$



Compressive Stress [Sc]:

$$= P/A_r$$

$$= 9123/77.2$$

$$= 118.111 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	275.0	...	38.5	...	0.243E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 275 - 14$$

$$= 261.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/77.2$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 275/2$$

$$= 137.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(2426/77.2)}$$

$$= 56.048 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1500 - \cos(120/2) (965 + 28 + 14) - 24$$

$$= 972.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038900/2651)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 972/56$$

$$= 17.351$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 262 / (2 \cdot 1800) \cdot 581 \cdot 972/2$$

$$= 20.580 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($17.4 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (17.4)^2 / (2 \cdot 123^2)) 2651 /$$

$$(5/3 + 3 \cdot (17.4) / (8 \cdot 123) - (17.4)^3 / (8 \cdot 123^3))$$

$$= 1527 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 118/1527 + (20.6 \cdot 138/24263022) / 1767$$

$$= 0.084$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	261.0	0.0	36.5	0.0	0.243E+04



Web	581.4	14.0	0.0	81.4	0.0	13.3
Totals	...	...	...	117.9	...	0.244E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/118$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 275/2$$

$$= 137.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1500 - \sqrt{(1007 + 14)^2 - (581/2)^2} - 24$$

$$= 511.871 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(2439/118)}$$

$$= 45.479 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 512/45.5$$

$$= 11.255$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 262/(2 \cdot 1800)$$

$$= 0.073 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.073 \cdot 581 \cdot 512$$

$$= 21.665 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 21.7/177402$$

$$= 12.212 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($11.3 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.3)^2 / (2 \cdot 123^2)) 2651 /$$

$$(5/3 + 3 \cdot (11.3) / (8 \cdot 123) - (11.3^3) / (8 \cdot 123^3))$$

$$= 1552 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 155/1552 + (21.7 \cdot 138/2439) / 1767$$

$$= 0.107$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BlArea	3.0418 cm ²
Saddle Load Q0 (Weight)	Q0	55310.6 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	70.7 kgf
Maximum Transverse Force	Ft	456.1 kgf
Maximum Longitudinal Force	Fl	262.1 kgf
Saddle Bolted to Steel Foundation	No	



Shear Stress in a Single Bolt, Longitudinal Direction [τ_b, l]:
= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $262 / (3.04 \cdot 4)$
= 21.5 kgf/cm^2 . Must be less than 1052.3 kgf/cm^2 .

Shear Stress in a Single Bolt, Transverse Direction [τ_b, t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $456 / (3.04 \cdot 4)$
= 37.5 kgf/cm^2 . Must be less than 1052.3 kgf/cm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [$Bltarearl$]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [$Bltarears$]:
= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $262 / (1052 \cdot 4)$
= 0.0623 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [$Rmom$]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1500 \cdot 456 + 0$
= 684.22 kgf-m .

Eccentricity (e):
= $Rmom / QO$
= $684 / 55311$
= $12.37 \text{ mm} < Bplen/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [$Bltareart$]:
= 0 (No Uplift)

Required Area of a Single Bolt [$Bltarear$]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0, 0.062, 0]$
= 0.0623 cm^2

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: A1-Tube Inlet

From: 30

Pressure for Reinforcement Calculations	P	10.197	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1930.00	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		100.82	cm
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		600.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		40	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	94.7929	mm
Height of Beveled Transition	L'	18.7929	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: A1-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	574.700	mm.
Actual Thickness Used in Calculation	17.450	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.2 \cdot 968 / (1407 \cdot 1 - 0.6 \cdot 10.2)$
 $= 7.0457 \text{ mm}$



Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.2 \cdot 290 / (1407 \cdot 1 - 0.6 \cdot 10.2)$
 $= 2.1133 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 32.6 / 0.5773$
 $= 56.3786 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1161.4008 mm
 Parallel to Vessel Wall, opening length d 580.7004 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 42.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407 / 1407)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407 / 1407)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	40.914	NA	NA
Area in Shell	A1	57.805	NA	NA
Area in Nozzle Wall	A2	10.486	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	27.668	NA	NA
TOTAL AREA AVAILABLE	Atot	96.959	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (581 \cdot 7.05 \cdot 1 + 2 \cdot 47 \cdot 7.05 \cdot 1 \cdot (1 - 1))$
 $= 40.914 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 581 (1 \cdot 17 - 1 \cdot 7.05) - 2 \cdot 47$
 $(1 \cdot 17 - 1 \cdot 7.05) \cdot (1 - 1)$
 $= 57.805 \text{ cm}^2$



Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2}$$

$$= (2 \cdot 42.5) (14.4 - 2.11) 1$$

$$= 10.486 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_{r2} + (W_i - c_{an}/0.707)^2 \cdot f_{r2}$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(42.5, 300, 94.8)) \cdot (50 - 17.4) \cdot 1$$

$$= 27.668 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 5.1133 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 5.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 10.0457 \text{ mm}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 10.0457 \text{ mm}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 5.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.3, \max(10, 5.5)]$$

$$= 10.0457 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(5.11, 10)$$

$$= 10.0457 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 17.450 = 15.269 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 86.6,	Allowable	: 1407.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3431.3 kgf/cm ²	Passed
Occasional	: 100.0,	Allowable	: 1871.5 kgf/cm ²	Passed
Shear	: 34.9,	Allowable	: 985.0 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 15.3$, $t_r = 2.11$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.17$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	4 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-104 $^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 20$, $t_r = 7.05$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.41$, Temp. Reduction = $47 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-48 $^\circ\text{C}$
Min Metal Temp. w/o impact per UG-20(f)	-29 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : -48 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 $^\circ\text{C}$



Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 10.20/52.01 = 0.196

Weld Size Calculations, Description: A1-Tube Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 17.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (40.9 - 57.8 + 2 \cdot 47 \cdot 1 \cdot (1 \cdot 17 - 7.05)) \cdot 1407)$
 $= \max(0, -10600.76) \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can})/707)^2 \cdot f_r2 \cdot S_v$
 $= (38.2 + 1 - 0 \cdot 1) \cdot 1407$
 $= 55094.38 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (38.2 + 0 + 1 + (16)) \cdot 1407$
 $= 77580.35 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (38.2 + 0 + 1 + 0 + (16)) \cdot 1407$
 $= 77580.35 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 675 \cdot 10 \cdot 0.49 \cdot 1407$
 $= 73074. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 314) \cdot (50 - 3) \cdot 0.7 \cdot 1407$
 $= 456460. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 675 \cdot (14 - 0) \cdot 0.74 \cdot 1407$
 $= 154499. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (73074 + 456460) = 529534 \text{ kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (73074 + 0 + 154499 + 0) = 227573 \text{ kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$



= (73074 + 154499 + 0) = 227573 kgf

Summary of Failure Path Calculations:

Path 1-1 = 529534 kgf, must exceed W = 0 kgf or W1 = 55094 kgf
 Path 2-2 = 227572 kgf, must exceed W = 0 kgf or W2 = 77580 kgf
 Path 3-3 = 227572 kgf, must exceed W = 0 kgf or W3 = 77580 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 17 kgf/cm²

The Drop for this Nozzle is : 60.8873 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 380.8873 mm

Input Echo, WRC297 Item 1, Description: A1-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1930.000	mm
Shell Thickness	20.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	574.700	mm
Nozzle Thickness	50.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	10.20	kgf/cm ²
Radial Load	P	-3671.16	kgf
Circumferential Shear	Vc	3671.16	kgf
Longitudinal Shear	Vl	2753.37	kgf
Circumferential Moment	Mc	5.72	kgf-m.
Longitudinal Moment	Ml	5.72	kgf-m.
Torsional Moment	Mt	6.61	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

= NozzleOD + 2 • 1.65 • √(Rmean(t - ca))
 = 674.7 + 2 • 1.65 • √(976.5(20.0 - 3.0))
 = 1099.882 mm

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1953.000 mm
Nozzle Outside Diameter	(d) =	674.700 mm
Vessel Thickness used	(T) =	17.000 mm
Nozzle Thickness used	(t) =	47.000 mm



T / t = 0.362
 d / t = 14.355

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.703$$

Nr/P = 0.052
 Mr/P = 0.089
 MØ/P = 0.028
 NØ/P = 0.076
 MrD/Mc = 0.256
 NrDL/Mc = 0.084
 MØd/Mc = 0.078
 NØDL/Mc = 0.091
 MrD/Ml = 0.078
 NrDL/Ml = 0.045
 MØD/Ml = 0.023
 NØDL/Ml = 0.087

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	96	96	96	96
Outplane Bending (P)	212	-212	212	-212
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	575	585	575	585
Outplane Stress Summary	884	470	885	469

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	65	65	65	65
Inplane Bending (P)	675	-675	675	-675
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-1	1	1	-1
Inplane Pressure Stress	287	287	287	287
Inplane Stress Summary	1027	-321	1030	-323

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	20	20	-20	-20
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	20	20	-20	-20

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1030	792	1033	794

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------



Outplane Membrane (P)	96	96	96	96
Outplane Bending (P)	212	-212	212	-212
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-1	1	1	-1
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	287	287	287	287
Outplane Stress Summary	595	173	598	170

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	65	65	65	65
Inplane Bending (P)	675	-675	675	-675
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-4	4	4	-4
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	575	585	575	585
Inplane Stress Summary	1311	-20	1321	-28

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-15	-15	15	15
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-15	-15	15	15

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1312	195	1321	201

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	96	96	96	96
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	69	82	69	82
Hoop Stress Summary	166	178	166	178

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	62	-62	62	-62
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0



Axial Pressure Stress	34	34	34	34
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Axial Stress Summary	101	-23	101	-23
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	7	7	-7	-7
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	7	7	-7	-7

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	167	202	167	203

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	96	96	96	96
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	69	82	69	82
Hoop Stress Summary	166	178	166	178

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	62	-62	62	-62
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	34	34	34	34
Axial Stress Summary	101	-23	102	-24

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-5	-5	5	5
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-5	-5	5	5

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	166	202	167	203

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	575	585	575	585	575	585	575	585
Circ. Pl (SUS)	96	96	96	96	65	65	65	65
Circ. Q (SUS)	212	-212	212	-212	671	-671	679	-679
Long. Pm (SUS)	287	287	287	287	287	287	287	287
Long. Pl (SUS)	65	65	65	65	96	96	96	96
Long. Q (SUS)	674	-674	676	-676	211	-211	213	-213
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	20	20	-20	-20	-15	-15	15	15
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	575.0	585.0	575.0	585.0	575.0	585.0	575.0	585.0
Pm+Pl (SUS)	672.2	682.2	672.2	682.2	640.9	650.8	640.9	650.8
Pm+Pl+Q (Total)	1028	792	1030	794	1311	195	1319	201

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	585.00	1407.13	Passed
Pm+Pl (SUS)	682.21	2110.70	Passed
Pm+Pl+Q (TOTAL)	1319.31	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	69	82	69	82	69	82	69	82
Circ. Pl (SUS)	96	96	96	96	96	96	96	96
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	34	34	34	34	34	34	34	34
Long. Pl (SUS)	3	3	3	3	3	3	3	3
Long. Q (SUS)	62	-62	62	-62	62	-62	62	-62
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-5	-5	5	5
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	69.0	82.0	69.0	82.0	69.0	82.0	69.0	82.0
Pm+Pl (SUS)	165.4	178.3	165.4	178.3	165.2	178.2	165.2	178.2
Pm+Pl+Q (Total)	165.7	203.5	165.7	203.5	165.4	203.2	165.4	203.2

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	82.00	1407.13	Passed
Pm+Pl (SUS)	178.35	2110.70	Passed
Pm+Pl+Q (TOTAL)	203.48	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite 28 Licensee: SL User
FileName : Ammonia condenser-AK
Nozzle Calcs.: A1-Tube Inlet

Nozl: 9 10:06am Aug 31,2026

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Input, Nozzle Desc: B1-Tube Inlet

From: 30

Pressure for Reinforcement Calculations	P	10.390	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1930.00	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		100.82	cm
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		600.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		40	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	94.7929	mm
Height of Beveled Transition	L'	18.7929	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: B1-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	574.700	mm.
Actual Thickness Used in Calculation	17.450	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.4 \cdot 968 / (1407 \cdot 1 - 0.6 \cdot 10.4)$
 $= 7.1795 \text{ mm}$



Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.4 \cdot 290 / (1407 \cdot 1 - 0.6 \cdot 10.4)$
 $= 2.1535 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 32.6 / 0.5773$
 $= 56.3786 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1161.4008 mm
 Parallel to Vessel Wall, opening length d 580.7004 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 42.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407 / 1407)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407 / 1407)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(\text{fr2}, \text{fr4})$
 $= \min(1, 1)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	41.692	NA	NA
Area in Shell	A1	57.027	NA	NA
Area in Nozzle Wall	A2	10.452	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	27.668	NA	NA
TOTAL AREA AVAILABLE	Atot	96.147	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot \text{tr} \cdot F + 2 \cdot \text{tn} \cdot \text{tr} \cdot F \cdot (1 - \text{fr1}))$ UG-37(c)
 $= (581 \cdot 7.18 \cdot 1 + 2 \cdot 47 \cdot 7.18 \cdot 1 \cdot (1 - 1))$
 $= 41.692 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot \text{tr}) - 2 \cdot \text{tn}(E1 \cdot t - F \cdot \text{tr}) \cdot (1 - \text{fr1})$
 $= 581 (1 \cdot 17 - 1 \cdot 7.18) - 2 \cdot 47$
 $(1 \cdot 17 - 1 \cdot 7.18) \cdot (1 - 1)$
 $= 57.027 \text{ cm}^2$



Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2}$$

$$= (2 \cdot 42.5) (14.4 - 2.15) 1$$

$$= 10.452 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_{r2} + (W_i - c_{an}/0.707)^2 \cdot f_{r2}$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(42.5, 300, 94.8)) \cdot (50 - 17.4) \cdot 1$$

$$= 27.668 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.1535 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 10.1795 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 10.1795 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(10.2, 5.5)]$$

$$= 10.1795 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(5.15, 10.2)$$

$$= 10.1795 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 17.450 = 15.269 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 88.5,	Allowable	: 1407.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3429.4 kgf/cm ²	Passed
Occasional	: 101.9,	Allowable	: 1871.5 kgf/cm ²	Passed
Shear	: 34.9,	Allowable	: 985.0 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 15.3, tr = 2.15, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.18$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 20, tr = 7.18, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.42$, Temp. Reduction = 45 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
--	--------



Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 10.39/52.01 = 0.200

Weld Size Calculations, Description: B1-Tube Inlet

Intermediate Calc. for nozzle/shell welds Tmin 17.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (41.7 - 57 + 2 • 47 • 1 •
(1 • 17 - 7.18))1407)
= max(0, -8589.89) kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (38.1 + 1 - 0 • 1) • 1407
= 55046.34 kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (38.1 + 0 + 1 + (16)) • 1407
= 77532.32 kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (38.1 + 0 + 1 + 0 + (16)) • 1407
= 77532.32 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 675 • 10 • 0.49 • 1407
= 73074. kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 314) • (50 - 3) • 0.7 • 1407
= 456460. kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.14/2.0) • 675 • (14 - 0) • 0.74 • 1407
= 154499. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (73074 + 456460) = 529534 kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (73074 + 0 + 154499 + 0) = 227573 kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)



= (73074 + 154499 + 0) = 227573 kgf

Summary of Failure Path Calculations:

Path 1-1 = 529534 kgf, must exceed W = 0 kgf or W1 = 55046 kgf
 Path 2-2 = 227572 kgf, must exceed W = 0 kgf or W2 = 77532 kgf
 Path 3-3 = 227572 kgf, must exceed W = 0 kgf or W3 = 77532 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 16.8 kgf/cm²

The Drop for this Nozzle is : 60.8873 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 380.8873 mm

Input Echo, WRC297 Item 1, Description: B1-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1930.000	mm
Shell Thickness	20.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	574.700	mm
Nozzle Thickness	50.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	10.39	kgf/cm ²
Radial Load	P	-3671.16	kgf
Circumferential Shear	Vc	3671.16	kgf
Longitudinal Shear	Vl	2753.37	kgf
Circumferential Moment	Mc	5.72	kgf-m.
Longitudinal Moment	Ml	5.72	kgf-m.
Torsional Moment	Mt	6.61	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

= NozzleOD + 2 • 1.65 • √(Rmean(t - ca))
 = 674.7 + 2 • 1.65 • √(976.5(20.0 - 3.0))
 = 1099.882 mm

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1953.000 mm
Nozzle Outside Diameter	(d) =	674.700 mm
Vessel Thickness used	(T) =	17.000 mm
Nozzle Thickness used	(t) =	47.000 mm



T / t = 0.362
 d / t = 14.355

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.703$$

Nr/P = 0.052
 Mr/P = 0.089
 MØ/P = 0.028
 NØ/P = 0.076
 MrD/Mc = 0.256
 NrDL/Mc = 0.084
 MØd/Mc = 0.078
 NØDL/Mc = 0.091
 MrD/Ml = 0.078
 NrDL/Ml = 0.045
 MØD/Ml = 0.023
 NØDL/Ml = 0.087

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	96	96	96	96
Outplane Bending (P)	212	-212	212	-212
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	586	596	586	596
Outplane Stress Summary	894	481	896	480

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	65	65	65	65
Inplane Bending (P)	675	-675	675	-675
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-1	1	1	-1
Inplane Pressure Stress	293	293	293	293
Inplane Stress Summary	1032	-315	1035	-318

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	20	20	-20	-20
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	20	20	-20	-20

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	1035	797	1038	799

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
--	-----------------------	----------------------	------------------------	-----------------------



Outplane Membrane (P)	96	96	96	96
Outplane Bending (P)	212	-212	212	-212
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-1	1	1	-1
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	293	293	293	293

Outplane Stress Summary	600	178	603	176
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	65	65	65	65
Inplane Bending (P)	675	-675	675	-675
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-4	4	4	-4
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	586	596	586	596

Inplane Stress Summary	1322	-8	1332	-17
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-15	-15	15	15
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-15	-15	15	15
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	1323	189	1332	196
------------------------	------	-----	------	-----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	96	96	96	96
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	71	83	71	83

Hoop Stress Summary	167	180	168	180
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	3	3	3	3
Axial Bending (P)	62	-62	62	-62
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0



Axial Pressure Stress	35	35	35	35
-----------------------	----	----	----	----

Axial Stress Summary	102	-23	102	-23
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	7	7	-7	-7
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	7	7	-7	-7

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	168	203	168	204

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	96	96	96	96
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	71	83	71	83
Hoop Stress Summary	167	180	168	180

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	62	-62	62	-62
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	35	35	35	35
Axial Stress Summary	101	-22	102	-23

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-5	-5	5	5
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-5	-5	5	5

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	168	202	168	204

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):



Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	586	596	586	596	586	596	586	596
Circ. Pl (SUS)	96	96	96	96	65	65	65	65
Circ. Q (SUS)	212	-212	212	-212	671	-671	679	-679
Long. Pm (SUS)	293	293	293	293	293	293	293	293
Long. Pl (SUS)	65	65	65	65	96	96	96	96
Long. Q (SUS)	674	-674	676	-676	211	-211	213	-213
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	20	20	-20	-20	-15	-15	15	15
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	586.0	596.0	586.0	596.0	586.0	596.0	586.0	596.0
Pm+Pl (SUS)	683.2	693.2	683.2	693.2	651.9	661.8	651.9	661.8
Pm+Pl+Q (Total)	1034	797	1036	799	1322	190	1330	196

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	596.00	1407.13	Passed
Pm+Pl (SUS)	693.19	2110.70	Passed
Pm+Pl+Q (TOTAL)	1330.31	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	71	83	71	83	71	83	71	83
Circ. Pl (SUS)	96	96	96	96	96	96	96	96
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	35	35	35	35	35	35	35	35
Long. Pl (SUS)	3	3	3	3	3	3	3	3
Long. Q (SUS)	62	-62	62	-62	62	-62	62	-62
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-5	-5	5	5
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	71.0	83.0	71.0	83.0	71.0	83.0	71.0	83.0
Pm+Pl (SUS)	167.4	179.3	167.4	179.3	167.2	179.2	167.2	179.2
Pm+Pl+Q (Total)	167.7	203.5	167.7	203.5	167.4	203.2	167.4	203.2

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	83.00	1407.13	Passed
Pm+Pl (SUS)	179.35	2110.70	Passed
Pm+Pl+Q (TOTAL)	203.48	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite 28 Licensee: SL User
FileName : Ammonia condenser-AK
Nozzle Calcs.: B1-Tube Inlet

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Input, Nozzle Desc: A2-Shell Inlet

From: 40

Pressure for Reinforcement Calculations	P	25.495	kgf/cm ²
Temperature for Internal Pressure	Temp	180	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1930.00	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		291.32	cm
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		360.00	deg
Diameter		250.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	96.1713	mm
Height of Beveled Transition	L'	20.1713	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: A2-Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	242.926	mm.
Actual Thickness Used in Calculation	15.062	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$\begin{aligned}
 &= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= 25.5 \cdot 966 / (1407 \cdot 1 - 0.6 \cdot 25.5) \\
 &= 17.7042 \text{ mm}
 \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned}
 &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= 25.5 \cdot 123 / (1407 \cdot 1 - 0.6 \cdot 25.5) \\
 &= 2.2524 \text{ mm}
 \end{aligned}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$\begin{aligned}
 &= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30) \\
 &= 34.9 / 0.5773 \\
 &= 60.5141 \text{ mm}
 \end{aligned}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	491.8512	mm
Parallel to Vessel Wall, opening length	d	245.9256	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	66.2500	mm

Weld Strength Reduction Factor [fr1]:

$$\begin{aligned}
 &= \min(1, S_n / S_v) \\
 &= \min(1, 1407 / 1407) \\
 &= 1.000
 \end{aligned}$$

Weld Strength Reduction Factor [fr2]:

$$\begin{aligned}
 &= \min(1, S_n / S_v) \\
 &= \min(1, 1407 / 1407) \\
 &= 1.000
 \end{aligned}$$

Weld Strength Reduction Factor [fr3]:

$$\begin{aligned}
 &= \min(fr2, fr4) \\
 &= \min(1, 1) \\
 &= 1.000
 \end{aligned}$$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	43.539	NA	NA
Area in Shell	A1	21.631	NA	NA
Area in Nozzle Wall	A2	14.985	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	46.293	NA	NA
TOTAL AREA AVAILABLE	Atot	83.909	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (246 \cdot 17.7 \cdot 1 + 2 \cdot 48.5 \cdot 17.7 \cdot 1 \cdot (1 - 1)) \\
 &= 43.539 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$



$$= 246 (1 \cdot 26.5 - 1 \cdot 17.7) - 2 \cdot 48.5 \\ (1 \cdot 26.5 - 1 \cdot 17.7) \cdot (1 - 1) \\ = 21.631 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2} \\ = (2 \cdot 66.3) (13.6 - 2.25) \cdot 1 \\ = 14.985 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_{r2} + (W_i - c_{an}/0.707)^2 \cdot f_{r2} \\ = 10^2 \cdot 1 + (0)^2 \cdot 1 \\ = 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2} \\ = (2 \cdot \min(66.3, 300, 96.2)) \cdot (50 - 15.1) \cdot 1 \\ = 46.293 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.7524 \text{ mm}$
Wall Thickness per UG16(b),	$t_{r16b} = 3.0000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$t_{rb1} = 19.2042 \text{ mm}$
Wall Thickness	$t_{b1} = \max(t_{rb1}, t_{r16b}) = 19.2042 \text{ mm}$
Wall Thickness	$t_{b2} = \max(t_{rb2}, t_{r16b}) = 3.0000 \text{ mm}$
Wall Thickness per table UG-45	$t_{b3} = 9.6026 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[t_{b3}, \max(t_{b1}, t_{b2})] \\ = \min[9.6, \max(19.2, 3)] \\ = 9.6026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b) \\ = \max(3.75, 9.6) \\ = 9.6026 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 15.062 = 13.179 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 95.9,	Allowable	: 1407.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3421.9 kgf/cm ²	Passed
Occasional	: 109.5,	Allowable	: 1871.5 kgf/cm ²	Passed
Shear	: 36.6,	Allowable	: 985.0 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 13.2$, $t_r = 2.25$, $c = 1.5 \text{ mm}$, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.19$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A	0°C
Min Metal Temp. at Required thickness (UCS 66.1)	-104°C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 28$, $t_r = 17.7$, $c = 1.5 \text{ mm}$, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.67$, Temp. Reduction = 18°C

Min Metal Temp. w/o impact per UCS-66, Curve B	2°C
Min Metal Temp. at Required thickness (UCS 66.1)	-16°C

Gov. MDMT of the nozzle to shell joint welded assembly : -16°C



Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 25.50/52.01 = 0.490

Weld Size Calculations, Description: A2-Shell Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (43.5 - 21.6 + 2 • 48.5 • 1 •
(1 • 26.5 - 17.7))1407)
= 42832.95 kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (61.3 + 1 - 0 • 1) • 1407
= 87633.47 kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (61.3 + 0 + 1 + (25.7)) • 1407
= 123803.81 kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (61.3 + 0 + 1 + 0 + (25.7)) • 1407
= 123803.81 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 343 • 10 • 0.49 • 1407
= 37141. kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 147) • (50 - 1.5) • 0.7 • 1407
= 220938. kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.14/2.0) • 343 • (24 - 0) • 0.74 • 1407
= 134616. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (37141 + 220938) = 258079 kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (37141 + 0 + 134616 + 0) = 171757 kgf



Failure Path [Path 3-3]:
 = (Sonw + Tngw + Sinw)
 = (37141 + 134616 + 0) = 171757 kgf

Summary of Failure Path Calculations:

Path 1-1 = 258078 kgf, must exceed W = 42832 kgf or W1 = 87633 kgf
 Path 2-2 = 171757 kgf, must exceed W = 42832 kgf or W2 = 123803 kgf
 Path 3-3 = 171757 kgf, must exceed W = 42832 kgf or W3 = 123803 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 36.7 kgf/cm²

The Drop for this Nozzle is : 15.3551 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 343.3551 mm

Input Echo, WRC297 Item 1, Description: A2-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1930.000	mm
Shell Thickness	28.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	242.926	mm
Nozzle Thickness	50.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUS tained loads.*

Design Internal Pressure	Dp	25.50	kgf/cm ²
Radial Load	P	-1529.65	kgf
Circumferential Shear	Vc	1529.65	kgf
Longitudinal Shear	Vl	1147.24	kgf
Circumferential Moment	Mc	0.99	kgf-m.
Longitudinal Moment	Ml	0.81	kgf-m.
Torsional Moment	Mt	1.15	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (6.859) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 = NozzleOD + 2 • 1.65 • √(Rmean(t - ca))
 = 342.926 + 2 • 1.65 • √(979.75(28.0 - 1.5))
 = 874.660 mm

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1959.500 mm
 Nozzle Outside Diameter (d) = 342.926 mm
 Vessel Thickness used (T) = 26.500 mm
 Nozzle Thickness used (t) = 48.500 mm
 T / t = 0.546
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.505$$

Nr/P = 0.083
 Mr/P = 0.170
 M0/P = 0.050
 N0/P = 0.111
 MrD/Mc = 0.293
 NrDL/Mc = 0.102
 M0d/Mc = 0.093
 N0DL/Mc = 0.057
 MrD/Ml = 0.144
 NrDL/Ml = 0.078
 M0D/Ml = 0.047
 N0DL/Ml = 0.131

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	24	24	24	24
Outplane Bending (P)	65	-65	65	-65
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	917	942	917	942
Outplane Stress Summary	1006	901	1006	901

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	18	18	18	18
Inplane Bending (P)	221	-221	221	-221
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	458	458	458	458
Inplane Stress Summary	698	255	698	254

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	10	10	-10	-10
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	10	10	-10	-10

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1007	901	1007	901



Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	24	24	24	24
Outplane Bending (P)	65	-65	65	-65
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	458	458	458	458
Outplane Stress Summary	547	417	548	417

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	18	18	18	18
Inplane Bending (P)	221	-221	221	-221
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	917	942	917	942
Inplane Stress Summary	1156	739	1157	738

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-8	-8	8	8
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-8	-8	8	8

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1156	740	1157	738

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	24	24	24	24
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	64	95	64	95
Hoop Stress Summary	89	119	89	119

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	3	3	3	3



Axial Bending (P)	49	-49	49	-49
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	32	32	32	32

Axial Stress Summary	85	-14	85	-14
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	5	5	-5	-5
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	5	5	-5	-5

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	93	134	93	134

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	24	24	24	24
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	64	95	64	95
Hoop Stress Summary	89	119	89	119

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	49	-49	49	-49
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	32	32	32	32
Axial Stress Summary	85	-13	85	-14

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-4	-4	4	4

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	91	133	91	134

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	917	942	917	942	917	942	917	942
Circ. Pl (SUS)	24	24	24	24	18	18	18	18
Circ. Q (SUS)	65	-65	65	-65	221	-221	221	-221
Long. Pm (SUS)	458	458	458	458	458	458	458	458
Long. Pl (SUS)	18	18	18	18	24	24	24	24
Long. Q (SUS)	221	-221	221	-221	65	-65	65	-65
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	10	10	-10	-10	-8	-8	8	8
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	917.0	942.0	917.0	942.0	917.0	942.0	917.0	942.0
Pm+Pl (SUS)	941.2	966.2	941.2	966.2	935.1	960.1	935.1	960.1
Pm+Pl+Q (Total)	1006	901	1006	901	1156	739	1156	739

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	942.00	1407.13	Passed
Pm+Pl (SUS)	966.20	2110.70	Passed
Pm+Pl+Q (TOTAL)	1156.11	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	64	95	64	95	64	95	64	95
Circ. Pl (SUS)	24	24	24	24	24	24	24	24
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	32	32	32	32	32	32	32	32
Long. Pl (SUS)	3	3	3	3	3	3	3	3
Long. Q (SUS)	49	-49	49	-49	49	-49	49	-49
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-4	-4	4	4
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	64.0	95.0	64.0	95.0	64.0	95.0	64.0	95.0
Pm+Pl (SUS)	88.5	119.3	88.5	119.3	88.3	119.2	88.3	119.2
Pm+Pl+Q (Total)	91.4	133.4	91.4	133.4	90.5	133.2	90.5	133.2

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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PV Elite 28 Licensee: SL User
FileName : Ammonia condenser-AK
Nozzle Calcs.: A2-Shell Inlet

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HEXAGON

Pm (SUS)	95.00	1407.13	Passed
Pm+Pl (SUS)	119.30	2110.70	Passed
Pm+Pl+Q (TOTAL)	133.38	4221.40	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: A3-Shell Inlet

From: 40

Pressure for Reinforcement Calculations	P	25.495	kgf/cm ²
Temperature for Internal Pressure	Temp	180	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1930.00	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1081.32	cm
User Entered Minimum Design Metal Temperature		-35.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		360.00	deg
Diameter		250.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	96.1713	mm
Height of Beveled Transition	L'	20.1713	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: A3-Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	242.926	mm.
Actual Thickness Used in Calculation	15.062	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$\begin{aligned}
 &= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= 25.5 \cdot 966 / (1407 \cdot 1 - 0.6 \cdot 25.5) \\
 &= 17.7042 \text{ mm}
 \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned}
 &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= 25.5 \cdot 123 / (1407 \cdot 1 - 0.6 \cdot 25.5) \\
 &= 2.2524 \text{ mm}
 \end{aligned}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$\begin{aligned}
 &= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30) \\
 &= 34.9 / 0.5773 \\
 &= 60.5141 \text{ mm}
 \end{aligned}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	491.8512	mm
Parallel to Vessel Wall, opening length	d	245.9256	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	66.2500	mm

Weld Strength Reduction Factor [fr1]:

$$\begin{aligned}
 &= \min(1, S_n / S_v) \\
 &= \min(1, 1407 / 1407) \\
 &= 1.000
 \end{aligned}$$

Weld Strength Reduction Factor [fr2]:

$$\begin{aligned}
 &= \min(1, S_n / S_v) \\
 &= \min(1, 1407 / 1407) \\
 &= 1.000
 \end{aligned}$$

Weld Strength Reduction Factor [fr3]:

$$\begin{aligned}
 &= \min(fr2, fr4) \\
 &= \min(1, 1) \\
 &= 1.000
 \end{aligned}$$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	43.539	NA	NA
Area in Shell	A1	21.631	NA	NA
Area in Nozzle Wall	A2	14.985	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	46.293	NA	NA
TOTAL AREA AVAILABLE	Atot	83.909	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (246 \cdot 17.7 \cdot 1 + 2 \cdot 48.5 \cdot 17.7 \cdot 1 \cdot (1 - 1)) \\
 &= 43.539 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$



$$= 246 (1 \cdot 26.5 - 1 \cdot 17.7) - 2 \cdot 48.5$$

$$(1 \cdot 26.5 - 1 \cdot 17.7) \cdot (1 - 1)$$

$$= 21.631 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2}$$

$$= (2 \cdot 66.3) (13.6 - 2.25) 1$$

$$= 14.985 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_{r2} + (W_i - c_n / 0.707)^2 \cdot f_{r2}$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(66.3, 300, 96.2)) \cdot (50 - 15.1) \cdot 1$$

$$= 46.293 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.7524 \text{ mm}$
Wall Thickness per UG16(b),	$t_{r16b} = 3.0000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$t_{rb1} = 19.2042 \text{ mm}$
Wall Thickness	$t_{b1} = \max(t_{rb1}, t_{r16b}) = 19.2042 \text{ mm}$
Wall Thickness	$t_{b2} = \max(t_{rb2}, t_{r16b}) = 3.0000 \text{ mm}$
Wall Thickness per table UG-45	$t_{b3} = 9.6026 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[t_{b3}, \max(t_{b1}, t_{b2})]$$

$$= \min[9.6, \max(19.2, 3)]$$

$$= 9.6026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.75, 9.6)$$

$$= 9.6026 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 15.062 = 13.179 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 95.9,	Allowable	: 1407.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3421.9 kgf/cm ²	Passed
Occasional	: 109.5,	Allowable	: 1871.5 kgf/cm ²	Passed
Shear	: 36.6,	Allowable	: 985.0 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 13.2$, $t_r = 2.25$, $c = 1.5 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.19$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A	0°C
Min Metal Temp. at Required thickness (UCS 66.1)	-104°C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 28$, $t_r = 17.7$, $c = 1.5 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.67$, Temp. Reduction = 18°C

Min Metal Temp. w/o impact per UCS-66, Curve B	2°C
Min Metal Temp. at Required thickness (UCS 66.1)	-16°C

Gov. MDMT of the nozzle to shell joint welded assembly : -16°C



Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 25.50/52.01 = 0.490

Weld Size Calculations, Description: A3-Shell Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (43.5 - 21.6 + 2 \cdot 48.5 \cdot 1 \cdot (1 \cdot 26.5 - 17.7)) \cdot 1407)$
 $= 42832.95 \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can})/0.707)^2 \cdot f_{r2} \cdot S_v$
 $= (61.3 + 1 - 0 \cdot 1) \cdot 1407$
 $= 87633.47 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (61.3 + 0 + 1 + (25.7)) \cdot 1407$
 $= 123803.81 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (61.3 + 0 + 1 + 0 + (25.7)) \cdot 1407$
 $= 123803.81 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 343 \cdot 10 \cdot 0.49 \cdot 1407$
 $= 37141. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 147) \cdot (50 - 1.5) \cdot 0.7 \cdot 1407$
 $= 220938. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 343 \cdot (24 - 0) \cdot 0.74 \cdot 1407$
 $= 134616. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (37141 + 220938) = 258079 \text{ kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (37141 + 0 + 134616 + 0) = 171757 \text{ kgf}$



Failure Path [Path 3-3]:
 = (Sonw + Tngw + Sinw)
 = (37141 + 134616 + 0) = 171757 kgf

Summary of Failure Path Calculations:

Path 1-1 = 258078 kgf, must exceed W = 42832 kgf or W1 = 87633 kgf
 Path 2-2 = 171757 kgf, must exceed W = 42832 kgf or W2 = 123803 kgf
 Path 3-3 = 171757 kgf, must exceed W = 42832 kgf or W3 = 123803 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 36.7 kgf/cm²

The Drop for this Nozzle is : 15.3551 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 343.3551 mm

Input Echo, WRC297 Item 1, Description: A3-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1930.000	mm
Shell Thickness	28.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	242.926	mm
Nozzle Thickness	50.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUS tained loads.*

Design Internal Pressure	Dp	25.50	kgf/cm ²
Radial Load	P	-1529.65	kgf
Circumferential Shear	Vc	1529.65	kgf
Longitudinal Shear	Vl	1147.24	kgf
Circumferential Moment	Mc	0.99	kgf-m.
Longitudinal Moment	Ml	0.99	kgf-m.
Torsional Moment	Mt	1.15	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (6.859) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 = NozzleOD + 2 • 1.65 • √(Rmean(t - ca))
 = 342.926 + 2 • 1.65 • √(979.75(28.0 - 1.5))
 = 874.660 mm

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1959.500 mm
 Nozzle Outside Diameter (d) = 342.926 mm
 Vessel Thickness used (T) = 26.500 mm
 Nozzle Thickness used (t) = 48.500 mm
 T / t = 0.546
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.505$$

Nr/P = 0.083
 Mr/P = 0.170
 M0/P = 0.050
 N0/P = 0.111
 MrD/Mc = 0.293
 NrDL/Mc = 0.102
 M0d/Mc = 0.093
 N0DL/Mc = 0.057
 MrD/Ml = 0.144
 NrDL/Ml = 0.078
 M0D/Ml = 0.047
 N0DL/Ml = 0.131

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	24	24	24	24
Outplane Bending (P)	65	-65	65	-65
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	917	942	917	942
Outplane Stress Summary	1006	901	1006	901

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	18	18	18	18
Inplane Bending (P)	221	-221	221	-221
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	458	458	458	458
Inplane Stress Summary	698	255	698	254

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	10	10	-10	-10
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	10	10	-10	-10

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1007	901	1007	901



Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	24	24	24	24
Outplane Bending (P)	65	-65	65	-65
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	458	458	458	458
Outplane Stress Summary	547	417	548	417

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	18	18	18	18
Inplane Bending (P)	221	-221	221	-221
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	917	942	917	942
Inplane Stress Summary	1156	739	1157	738

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-8	-8	8	8
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-8	-8	8	8

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1156	740	1157	738

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	24	24	24	24
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	64	95	64	95
Hoop Stress Summary	89	119	89	119

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	3	3	3	3



Axial Bending (P)	49	-49	49	-49
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	32	32	32	32

Axial Stress Summary	85	-14	85	-14
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	5	5	-5	-5
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	5	5	-5	-5

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	93	134	93	134

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	24	24	24	24
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	64	95	64	95
Hoop Stress Summary	89	119	89	119

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	49	-49	49	-49
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	32	32	32	32
Axial Stress Summary	85	-13	85	-14

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-4	-4	4	4

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	91	133	91	134



WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	917	942	917	942	917	942	917	942
Circ. Pl (SUS)	24	24	24	24	18	18	18	18
Circ. Q (SUS)	65	-65	65	-65	221	-221	221	-221
Long. Pm (SUS)	458	458	458	458	458	458	458	458
Long. Pl (SUS)	18	18	18	18	24	24	24	24
Long. Q (SUS)	221	-221	221	-221	65	-65	65	-65
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	10	10	-10	-10	-8	-8	8	8
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	917.0	942.0	917.0	942.0	917.0	942.0	917.0	942.0
Pm+Pl (SUS)	941.2	966.2	941.2	966.2	935.1	960.1	935.1	960.1
Pm+Pl+Q (Total)	1006	901	1006	901	1156	739	1156	739

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	942.00	1407.13	Passed
Pm+Pl (SUS)	966.20	2110.70	Passed
Pm+Pl+Q (TOTAL)	1156.11	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	64	95	64	95	64	95	64	95
Circ. Pl (SUS)	24	24	24	24	24	24	24	24
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	32	32	32	32	32	32	32	32
Long. Pl (SUS)	3	3	3	3	3	3	3	3
Long. Q (SUS)	49	-49	49	-49	49	-49	49	-49
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-4	-4	4	4
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	64.0	95.0	64.0	95.0	64.0	95.0	64.0	95.0
Pm+Pl (SUS)	88.5	119.3	88.5	119.3	88.3	119.2	88.3	119.2
Pm+Pl+Q (Total)	91.4	133.4	91.4	133.4	90.5	133.2	90.5	133.2

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	95.00	1407.13	Passed
Pm+Pl (SUS)	119.30	2110.70	Passed
Pm+Pl+Q (TOTAL)	133.38	4221.40	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main shell-1

Shell Maximum Design Pressure	Psd,max	25.49	kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	25.49	kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	0.00	kgf/cm ²
Shell Thickness	ts	28.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	casext	0.0000	mm
Inside Diameter of Shell	Ds	1930.000	mm
Shell Circumferential Joint Efficiency	Esw	1.000	
Shell Temperature for Internal Pressure	Ts	180.00	°C
Shell Material		SA-516 70	

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700	
Shell Allowable Stress at Temperature	Ss	1407.13	kgf/cm ²
Shell Allowable Stress at Ambient		1407.13	kgf/cm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder	
Channel Maximum Design Pressure	Ptd,max	10.20	kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	10.20	kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00	kgf/cm ²
Channel Thickness	tc	20.0000	mm
Channel Internal Corrosion Allowance	cac	3.0000	mm
Channel External Corrosion Allowance	cacext	0.0000	mm
Inside Diameter of Channel	Dc	1930.000	mm
Channel Design Temperature	TEMPC	85.00	°C
Channel Flange Matl		SA-516 70	

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700	
Channel Allowable Stress at Temperature	Sc	1407.13	kgf/cm ²
Channel Allowable Stress at Ambient		1407.13	kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	4722	
Tube Wall Thickness	et	2.1000	mm
Tube Outside Diameter	D	19.0500	mm
Total Straight Tube Length	Lt	9970.00	mm
Straight Tube Length (bet. inner tubsht faces) L		9260.00	mm
Design Temperature of the Tubes		180.00	°C
Tube Material		SA-334 6	
Tube Material UNS Number		K03006	
Is this a Welded Tube		No	
Tube Material Specification used	Smls. & wld. tube		
Tube Allowable Stress at Temperature		1203.20	kgf/cm ²
Tube Allowable Stress At Ambient		1203.20	kgf/cm ²
Tube Yield Stress At design Temperature	Syt	2135.19	kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	25.4000	mm
Tube Layout Pattern		Triangular	
Fillet Weld Leg	af	1.3000	mm
Groove Weld Leg	ag	1.3000	mm
Tube-Tubesheet Joint Weld Type		Full Strength	
Method for Tube-Tubesheet Jt. Allowable		UW-20.3	
Is the Tube Inset		No	
Radius to Outermost Tube Hole Center	ro	940.081	mm
Largest Center-to-Center Tube Distance	Ul	0.0000	mm
Length of Expanded Portion of Tube	ltx	0.0000	mm
Tube-side pass partition groove depth	hg	6.0000	mm
Area of Tube Layout	Ap	26284.4	cm ²
Perimeter of Tube Layout	Cp	6032.4	mm



Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature T 180.00 °C
 Tubesheet Material SA-350 LF2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03011
 Tubesheet Allowable Stress at Temperature S 1407.13 kgf/cm²
 Tubesheet Allowable Stress at Ambient Tt 1407.13 kgf/cm²
 Thickness of Tubesheet h 355.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 1.5000 mm
 Tubesheet Corr. Allowance (Channel side) Catc 3.0000 mm
 Tubesheet Outside Diameter A 1986.000 mm

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 470.000 mm
 Tube End condition corresponding to Span (l) k 1.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes

Mean Shell Metal Temp. along Shell len. Tsm 0.00 °C
 Mean Tube Metal Temp. along Tube length Ttm 66.00 °C
 Junction Stress Reduction option Perform Plastic Calculation

Tubesheet Integral with Both
 Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	180.0 °C	0.19739E+07 kgf/cm ²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20812E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	85.0 °C	0.20297E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	66.0 °C	0.0000118819 /°C
Elastic Mod. at Design Temperature	180.0 °C	0.19739E+07 kgf/cm ²
Elastic Mod. at Metal Temp. along Len	66.0 °C	0.20408E+07 kgf/cm ²
Elastic Mod. at Tubsht. Design Temp.	180.0 °C	0.19739E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%



Elastic Mod. at Design Temperature 180.0 °C 0.19739E+07 kgf/cm²
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20670E+07 kgf/cm²

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (10.2 \cdot (19/2 - 0)) / (1203 \cdot 1 + 0.4 \cdot 10.2)$
 $= 0.0805 + 0.0000 = 0.0805 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 180.00 °C
 Elastic Modulus for Material 1999551.12 kgf/cm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1000	19.05	9260.00	9.07	50.0000	0.0133672	1222.02

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 190 \text{ kgf/cm}^2$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.4940	19.05	9260.00	38.56	50.0000	0.0007398	737.36

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 737) / (3 \cdot 38.6) = 25.5 \text{ kgf/cm}^2$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.4940 mm
Allowable Internal Pressure at Corroded thickness	290.93 kgf/cm ²
Required Internal Design Pressure	10.20 kgf/cm ²
Allowable External Pressure at Corroded thickness	190.12 kgf/cm ²
Required External Design Pressure	25.49 kgf/cm ²
Required Thickness due to Shell Side pressure	0.4940 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [ρ]:

$= l_{tx} / h$ (modified for corrosion if present)
 $= 0/355$
 $= 0.0000$ (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter [d*]:

$= \max(dt - 2 \cdot tt(Et/E) \cdot (St/S) \cdot (\rho); dt - 2 \cdot tt)$
 $= \max(19 - 2 \cdot 2.1 (1973947/1973947) \cdot (1203/1407) \cdot (0); 19 - 2 \cdot 2.1)$
 $= 19.0500 \text{ mm}$

Equivalent outer tube limit circle diameter [Do]:

$= 2 \cdot ro + dt$
 $= 2 \cdot 940 + 19$
 $= 1899.212 \text{ mm}$

Determine the basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (25.4 - 19) / 25.4$
 $= 0.250$



Equivalent outer tube limit radius [ao]:

$$= Do / 2$$

$$= 1899/2$$

$$= 949.6060 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{(1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))}$$

$$= 25.4 / \sqrt{(1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 1899 \cdot 25.4) / (\pi \cdot 1899^2))}$$

$$= 25.4000 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (25.4 - 19) / 25.4$$

$$= 0.2500$$

Ratio [ps]:

$$= as / ao$$

$$= 965/950$$

$$= 1.0162$$

Ratio [pc]:

$$= ac / ao$$

$$= 965/950$$

$$= 1.0162$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 4722 ((19 - 2 \cdot 2.1) / (2 \cdot 950))^2$$

$$= 0.7113$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 4722 (19 / (2 \cdot 950))^2$$

$$= 0.5249$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((6 - 0); 0)$$

$$= 6.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts(Ds + ts)Es/L$$

$$= \pi \cdot 28 (1930 + 28)1973947/9260$$

$$= 367150.9375 \text{ kgf/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt(Dt - tt)Et/L$$

$$= \pi \cdot 2.1 (19 - 2.1)1973947/9260$$

$$= 238.3762 \text{ kgf/mm}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 367151 / (4722 \cdot 238)$$

$$= 0.32618$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks/Kj)$$

$$= 1 / (1 + 367151/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{(Ds + ts)ts}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(1930 + 28)28}$$

$$= 0.0078 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot Es \cdot ts^3 / (6(1 - \nu_s^2))$$



$$= 0.0078 \cdot 1973947 \cdot 28^3 / (6(1 - 0.3^2))$$

$$= 61615112.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1930 + 28) \cdot 61615112 / 355^3 \cdot (1 + 355 \cdot 0.0078 + (355 \cdot 0.0078)^2 / 2)$$

$$= 122223.9375 \text{ kgf/cm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 1930^2 / (4 \cdot 1973947 \cdot 28) \cdot (1 - 1930 \cdot 0.3 / (2(1930 + 28)))$$

$$= 0.14357\text{E-}01 \text{ mm/kgf/cm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(1930 + 20)20}$$

$$= 0.0092 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_c^2))$$

$$= 0.0092 \cdot 2029713 \cdot 20^3 / (6(1 - 0.3^2))$$

$$= 27375082.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(1930 + 20 / \cos(0)) \cdot 27375082 / 355^3 \cdot (1 + 355 \cdot 0.0092 \cdot (1 + 1) / 2 + (355 \cdot 0.0092)^2 / 2 \cdot 1)$$

$$= 68776.7 \text{ kgf/cm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1930^2 / (4 \cdot 2029713 \cdot 20) \cdot (1 - 1930 \cdot 0.3 / (2(1930 + 20)))$$

$$= 0.19534\text{E-}01 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 13.976378 ; \mu^* = 0.250000 ; E^* = 427577 \text{ kgf/cm}^2$$

$$E^*/E = 0.216611 ; \nu^* = 0.396457$$

Note: As h/p (13.976) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$= ((24(1 - \nu^{*2})Nt \cdot Et \cdot t_t(d_t - t_t)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.4^2)4722 \cdot 1973947 \cdot 2.1(19 - 2.1)950^2) / (427578 \cdot 9260 \cdot 355^3))^{1/4}$$

$$= 2.4176$$

Values from Table 4.18.3

$$Z_d = 0.098832 ; Z_v = 0.135038 ; Z_m = 0.585290$$

$$Z_a = 0.121971\text{E+}01 ; Z_w = 0.135038$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1986 / 1899$$

$$= 1.0457$$

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.4) / 427578 \cdot (122224 + 68777 + 1973947 \cdot \ln(1.05))$$

$$= 0.3941$$



Compute parameter [φ]:

$$= (1 + v \cdot F) \\ = (1 + 0.4) \cdot 0.39 \\ = 0.5504$$

Compute parameter [Q1]:

$$= (p_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\ = (1.02 - 1 - 0.55 \cdot 0.14) / (1 + 0.55 \cdot 0.59) \\ = -0.043950$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot Xa^4 \\ = (0.099 + -0.044 \cdot 0.14) / 2 \cdot 2.42^4 \\ = 1.5869$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot Xa^4 \\ = (0.14 + -0.044 \cdot 0.59) / 2 \cdot 2.42^4 \\ = 1.8673$$

Compute parameter [U]:

$$= (Z_w + (p_s - 1) \cdot Z_m) \cdot Xa^4 / (1 + \phi \cdot Z_m) \\ = (0.14 + (1.02 - 1) \cdot 0.59) \cdot 2.42^4 / (1 + 0.55 \cdot 0.59) \\ = 3.7346$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

$$= 0 \text{ (config a)}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ws]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s) \\ = 1.02 \cdot 61615112 \cdot 0.0078 \cdot 0.01435736 (1 + 355 \cdot 0.0078) \\ = 26215.3555 \text{ mm}^2$$

Calculate parameter [ω*s]:

$$= a o^2 (p_s^2 - 1) \cdot (p_s - 1) / 4 - \omega_s \\ = 950^2 (1.02^2 - 1) \cdot (1.02 - 1) / 4 - 26215 \\ = -26095.9102 \text{ mm}^2$$

Parameter [wc]:

$$= p_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \\ \zeta c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)] \\ = 1.02 \cdot 27375082 [0.019534132 (1 + 0.0092 \cdot 1 \cdot 355) - \\ 0.000 (1/355 + 1 \cdot 0.0092/2)] \\ = 21348.0332 \text{ mm}^2$$

Parameter [ω*c]:

$$= a o^2 [(p_c^2 + 1) \cdot (p_c - 1) / 4 - (p_s - 1) / 2] - \omega_c \\ = 950^2 [(1.02^2 + 1) \cdot (1.02 - 1) / 4 - (1.02 - 1) / 2] - 21348 \\ = -21228.5840 \text{ mm}^2$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/Kst \cdot (D_s/D_o)^2] v_s - \\ [(p_s^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(D_j^2 - (D_s)^2)/D_o^2] \} \\ = 25.5 \{ 0.52 + 2(1 - 0.52) \cdot 0.3 + [2/0.33 \cdot \\ (1930/1899)^2] \cdot 0.3 - [(1.02^2 - 1)/(1 \cdot 0.33)] - \\ [(1 - 1)/(2 \cdot 1 \cdot 0.33)] [(0^2 - (1930)^2)/1899^2] \}$$



$$= 66.5219 \text{ kgf/cm}^2$$

Pressure [P't]:

$$\begin{aligned} &= [xt + 2(1 - xt)vt + 1/(J \cdot Kst)]Pt \\ &= [0.71 + 2(1 - 0.71)0.3 + 1/(1 \cdot 0.33)]10.2 \\ &= 40.2831 \text{ kgf/cm}^2 \end{aligned}$$

Pressure [P_γ]:

$$\begin{aligned} &= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2) \\ &= 4722 \cdot 238 \cdot 0 / (3.14 \cdot 950^2) \\ &= 0.0000 \text{ kgf/cm}^2 \end{aligned}$$

Pressure [P_w]:

$$\begin{aligned} &= -\gamma b \cdot U \cdot W^* / (2 \cdot \pi \cdot ao^2) \\ &= -0 \cdot 3.73 \cdot 0 / (2 \cdot \pi \cdot 950^2) \\ &= -0.0000 \text{ kgf/cm}^2 \end{aligned}$$

Calculate the pressure [Prim]:

$$\begin{aligned} &= -(U/ao^2) \cdot (\omega^*s \cdot Ps - \omega^*c \cdot Pt) \\ &= -(3.73/950^2) \cdot (-26096 \cdot 25.5 - -21229 \cdot 10.2) \\ &= 1.8587 \text{ kgf/cm}^2 \end{aligned}$$

Calculate the pressure [P_w]:

$$\begin{aligned} &= (U/ao^2) \cdot (\omega^*s \cdot P^*s - \omega^*c \cdot P^*c) \\ &= (3.73/950^2) \cdot (26215 \cdot 0 - 21348 \cdot 0) \\ &= 0.0000 \text{ kgf/cm}^2 \end{aligned}$$

Determine the effective pressure [Pe]:

$$\begin{aligned} &= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1)Qz2)) \cdot \\ &\quad (P^*s - P^*t + P_\gamma + P_w + P_w + Prim) \\ &= 0.1000E+01 \cdot 0.33 / (1 + 1 \cdot 0.33(1.59 + (1.02 - \\ &\quad 1)1.87)) \cdot (66.5 - 40.3 + 0 + 0 + -0 + 1.86) \\ &= 6.0000 \text{ kgf/cm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$\begin{aligned} &= [((\omega^*s \cdot Ps - \omega^*c \cdot Pt) - (\omega^*s \cdot P^*s - \omega^*c \cdot P^*c)) + \\ &\quad W^* \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm) \\ &= [((-26096 \cdot 25.5 - -21229 \cdot 10.2) - \\ &\quad (26215 \cdot 0 - 21348 \cdot 0)) \cdot 0.01 + \\ &\quad 0 \cdot 0 / (2 \cdot \pi)] / (1 + 0.55 \cdot 0.59) \\ &= -3394.5508 \text{ kgf} \end{aligned}$$

Calculate factor [Q3]:

$$\begin{aligned} &= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2) \\ &= -0.044 + 2 \cdot -3395 / (6 \cdot 950^2) \\ &= -0.16943 \end{aligned}$$

Fm Value from Table 4.18.3 = 0.089175

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned} &= (1.5 \cdot Fm / \mu^*) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe \\ &= (1.5 \cdot 0.089 / 0.25) \cdot (2 \cdot 950 / (355 - 6))^2 \cdot 6 \\ &= 95.0691 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 1407 = 2111 \text{ kgf/cm}^2$$

Tubesheet Bending Stress - final thickness [σf]:

$$\begin{aligned} &= (1.5 \cdot Fm / \mu^*) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe \\ &= (1.5 \cdot 0.031 / 0.25) \cdot (2 \cdot 950 / (40.1 - 6))^2 \cdot 3.64 \\ &= 2110.5259 \text{ kgf/cm}^2 \end{aligned}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$\begin{aligned} &= h + Cats + Catc \\ &= 40.1 + 0 + 0 \\ &= 40.1194 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:



Shear Stress check [Tlimit]:

$$\begin{aligned} &= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y) \\ &= (2 \cdot 0.25 \cdot 355/950) \cdot 1126 \\ &= 210.4168 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress is not required to be computed; [Pe] <= Taulimit

Could not converge on a minimum required tubesheet thk. to satisfy the Shear Stress criteria.



Use the best Estimated Required Tubesheet Thickness value as design guidance & Compare the Actual Stresses versus Allowable Stresses.

$$\begin{aligned} \text{Tubesheet thickness} &= 9.2895 \text{ mm} \\ \text{Tubesheet Shear Stress} &= 345.9966 \text{ kgf/cm}^2 \end{aligned}$$

Required Tubesheet Thickness for Given Loadings (Including CA) [H reqd]:

$$\begin{aligned} &= \max(H_{reqB}, H_{reqS}) \\ &= \max(40.1, 9.29) \\ &= 40.1194 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$F_{tmin} = 0.57, F_{tmax} = 1.3$$

First Extreme Tube Axial Stress from among all the tubes [σ_1]:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s) \\ &= ((25.5 \cdot 0.52 - 10.2 \cdot 0.71) - (6) \cdot 0.57) / \\ &\quad (0.71 - 0.52) \\ &= 14.4831 \text{ kgf/cm}^2 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_2]:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s) \\ &= ((25.5 \cdot 0.52 - 10.2 \cdot 0.71) - (6) \cdot 1.3) / \\ &\quad (0.71 - 0.52) \\ &= -8.8847 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(| \sigma_{t,1} |; | \sigma_{t,2} |) = 14.5 \text{ kgf/cm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 1203.2018 \text{ kgf/cm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$\begin{aligned} &= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w) X_{a^4}); 1.25) \\ &= \max((3.25 - 0.25(0.099 + -0.17 \cdot 0.14) 2.42^4); 1.25) \\ &= 2.0000 \text{ (Should be } \leq 2 \text{)} \end{aligned}$$

Determine the factor [rt]:

$$\begin{aligned} &= (\sqrt{ dt^2 + (dt - 2 \cdot tt)^2 }) / 4 \\ &= (\sqrt{ 19^2 + (19 - 2 \cdot 2.1)^2 }) / 4 \\ &= 6.0385 \text{ mm} \end{aligned}$$

Determine the factor [Ct]:

$$\begin{aligned} &= \sqrt{ (2 \cdot \pi^2 \cdot E / S_{yt}) } \\ &= \sqrt{ (2 \cdot \pi^2 \cdot 1973947 / 2135) } \\ &= 135.0872 \end{aligned}$$

Determine the factor [Ft]:

$$\begin{aligned} &= k \cdot L / r \\ &= 1 \cdot 470 / 6.04 \end{aligned}$$



$$= 77.8333$$

Buckling Allowable Stress [Stb]:

$$\begin{aligned} &= (S_y/t/F_s) \cdot (1 - F_t/(2 \cdot C_t)) \\ &= (2135/2) \cdot (1 - 77.8/(2 \cdot 135)) \\ &= 760.037 \text{ kgf/cm}^2 \quad (\text{Never greater than } S_{ot}) \end{aligned}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$\begin{aligned} &= \sigma_{t,max} \cdot \text{Tube Area} \\ &= 14.5 \cdot 1.12 \\ &= 16.20 \text{ kgf} \end{aligned}$$

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (19 - 2.1) 2.1 \\ &= 1.118 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S \\ &= \pi \cdot 2.1 (19 - 2.1) 1203 \\ &= 1345.478 \text{ kgf} \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \quad (\text{but not } > F_t) \\ &= 0.85 \cdot \pi \cdot 1.3 (19 + 0.67 \cdot 1.3) 1203 \\ &= 832.0724 \text{ kgf} \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned} &= S / S_w \quad \text{where } S_w = \min(S \text{ and } S_t) \\ &= 1203/1203 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned} &= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \quad (\text{but not } > F_t) \\ &= 0.55 \cdot \pi \cdot 1.3 (19 + 0.67 \cdot 1.3) 1203 \\ &= 538.3998 \text{ kgf} \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned} &= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right) \\ &= 2 \left(\sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 2.1 \cdot (19 - 2.1) 1 \cdot 1)} - 0.75 \cdot 19 \right) \\ &= 2.552 \text{ mm} \end{aligned}$$

$$\text{Minimum Required Fillet Weld Leg [afr]:} \quad 1.2759 \text{ mm}$$

$$\text{Minimum Required Groove Weld Leg [agr]:} \quad 1.2759 \text{ mm}$$

Tube-Tubesheet Jt allowable, 1345 is $\geq$ tube strength 1345 kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= S_s \cdot E_{sw} = 1407 \cdot 1 = 1407 \text{ kgf/cm}^2$$

Axial Membrane Stress in Shell [σ_s, m]:

$$\begin{aligned} &= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s) \\ &= 950^2 / ((1930 + 28) 28) \cdot [6 + (1.02^2 - 1) \\ &\quad (25.5 - 10.2)] + 965^2 \cdot 10.2 / ((1930 + 28) 28) \\ &= 280.1216 \text{ kgf/cm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:



Note:

For a given Shell thickness of 28 mm, the minimum Shell length adjacent to the tubesheet should be 418.437 mm

Shell Membrane Stress due to Joint Interaction $[\sigma_s, m]$:

$$= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1) (P_s - P_t)] + a_s^2 P_t / ((D_s + t_s) t_s)$$

$$= 950^2 / ((1930 + 28) 28) [6 + (1.02^2 - 1) (25.5 - 10.2)] + 965^2 \cdot 10.2 / ((1930 + 28) 28)$$

$$= 280.1216 \text{ kgf/cm}^2$$

Shell Bending Stress due to Joint Interaction $[\sigma_s, b]$:

$$= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s P_s + a_s^2 P_s / (E_s t_s)] + 6(1 - \nu_s^2) / (E_s) (a_o / h)^3 (1 + h \beta_s / 2) [P_e (Z_v + Z_m Q_1) + 2 / a_o^2 Z_m Q_2] \}$$

$$= 6 \cdot 61615112 / 28^2 \{ 0.0078 [0.01435736 \cdot 25.5 + 965^2 \cdot 0 / (55270528)] + 6(1 - 0.4^2) / (427578) (950 / 355)^3 (1 + 355 \cdot 0.0078 / 2) [6(0.14 + 0.59 \cdot -0.044) + 2 / 950^2 \cdot 0.59 \cdot -3395] \}$$

$$= 1394.6089 \text{ kgf/cm}^2$$

Shell Stress Summation vs. Allowable

$$:: |\sigma_s, m| + |\sigma_s, b| \leq 1.5 \cdot S_s$$

$$:: |280| + |1395| \leq 2111 \text{ kgf/cm}^2$$

$$:: 1675, \text{ must be } \leq 2111 \text{ kgf/cm}^2$$

Note:

For a given Channel thickness of 20 mm, the minimum Channel length adjacent to the tubesheet should be 353.644 mm

Channel Membrane Stress due to Joint Interaction $[\sigma_c, m]$:

$$= a_c^2 \cdot P_t / ((D_c + t_c) t_c \cdot \cos(\theta))$$

$$= 965^2 \cdot 10.2 / ((1930 + 20) 20 \cdot \cos(0))$$

$$= 243.489 \text{ kgf/cm}^2$$

Channel Bending Stress due to Joint Interaction $[\sigma_c, b]$:

$$= 6 \cdot k_c / t_c^2 \cdot \{ (\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6(1 - \nu_c^2) / E_c \cdot (a_o / h)^3 \cdot (C_m \theta + h \beta_c / 2 \cdot C_m \Delta) \cdot [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 27375082 / 20^2 \cdot \{ (0.019534132 \cdot 1 \cdot 0.0092 - 0.000 \cdot 1) \cdot 10.2 - 6(1 - 0.4^2) / 427578 \cdot (950 / 355)^3 \cdot (1 + 355 \cdot 0.0092 / 2 \cdot 1) \cdot [6(0.14 + 0.59 \cdot -0.044) + 2 / 950^2 \cdot 0.59 \cdot -3395] \}$$

$$= 700.2348 \text{ kgf/cm}^2$$

Channel Stress Summation vs. Allowable

$$:: |\sigma_c, m| + |\sigma_c, b| \leq 1.5 \cdot S_c$$

$$:: |243| + |700| \leq 2111 \text{ kgf/cm}^2$$

$$:: 944 \text{ must be } \leq 2111 \text{ kgf/cm}^2$$

Computations completed for ASME tubesheet configuration: a

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	95.1 <=	2110.7 kgf/cm ²	Pass
Tubesheet Shear Stress	29.5 <=	1125.7 kgf/cm ²	Pass
Maximum Tube Stress	14.5 <=	1203.2 kgf/cm ²	Pass
Minimum Tube Stress (Buckling)	-8.9 <=	-760.0 kgf/cm ²	Pass
Maximum Force on any one Tube	16.2 <=	1345.5 kgf	Pass
Axial Membrane Stress in Shell	280.1 <=	1407.1 kgf/cm ²	Pass
Shell Stress (jt. inter.)	1674.7 <=	2110.7 kgf/cm ²	Pass
Channel Stress (jt. inter.)	943.7 <=	2110.7 kgf/cm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
----------------	----------	--------	-----



Tubesheet Thickness :	40.119	355.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.276	1.300	Ok
Tube-Tubesheet Groove Weld Leg :	1.276	1.300	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Shear	Allwd	Case Type	Pass/ Fail
D1uc	42.557	...	177	2111	43	1126	Ps+Pt-Th	D1 Ok
D2uc	65.212	...	272	2111	73	1126	Ps+Pt-Th	D2 Ok
D3uc	40.119	...	95	2111	29	1126	Ps+Pt-Th	D3 Ok
O1uc	60.976	...	1127	4393	254	1126	Ps+Pt+Th	O1 Ok
O2uc	83.235	...	1561	4393	370	1126	Ps+Pt+Th	O2 Ok
O3uc	76.247	...	1390	4393	327	1126	Ps+Pt+Th	O3 Ok
O4uc	68.709	...	1298	4393	297	1126	Ps+Pt+Th	O4 Ok
D1c	40.824	...	180	2111	44	1126	Ps+Pt-Th-c	D1 Ok
D2c	63.371	...	274	2111	73	1126	Ps+Pt-Th-c	D2 Ok
D3c	40.331	...	94	2111	29	1126	Ps+Pt-Th-c	D3 Ok
O1c	60.333	...	1097	4393	245	1126	Ps+Pt+Th-c	O1 Ok
O2c	81.177	...	1543	4393	361	1126	Ps+Pt+Th-c	O2 Ok
O3c	74.906	...	1366	4393	318	1126	Ps+Pt+Th-c	O3 Ok
O4c	67.348	...	1274	4393	288	1126	Ps+Pt+Th-c	O4 Ok
Max:	83.2345	... mm		0.355		0.328	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	----	Shell Stresses	----	:	----	Shell Band Stress	----	:	Pass
	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Fail
D1uc	23	1407	...	...	:	...	...	...	Ok
D2uc	257	1407	...	...	:	...	...	...	Ok
D3uc	280	1407	...	...	:	...	...	...	Ok
O1uc	1018	4637	...	...	:	...	...	...	Ok
O2uc	1252	4637	...	...	:	...	...	...	Ok
O3uc	1275	4637	...	...	:	...	...	...	Ok
O4uc	995	4637	...	...	:	...	...	...	Ok
D1c	25	1407	...	...	:	...	...	...	Ok
D2c	271	1407	...	...	:	...	...	...	Ok
D3c	296	1407	...	...	:	...	...	...	Ok
O1c	1031	4637	...	...	:	...	...	...	Ok
O2c	1277	4637	...	...	:	...	...	...	Ok
O3c	1302	4637	...	...	:	...	...	...	Ok



04c 1006 4637 Ok

Max RATIO 0.281

Tube, Shell and Channel Stress Summary:

		---- Tube Stresses				Tube Loads		Shell Stress		Channel Stress		Stress Pass	
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd		Stress	Allwd	Stress	Allwd	Fail	
D1uc	31	1203	-20	-760	35	1345		213	2110	1180	2110	Ok	
D2uc	40	1203	-40	-760	45	1345		1842	2110	236	2110	Ok	
D3uc	14	1203	-9	-760	16	1345		1675	2110	944	2110	Ok	
O1uc	484	2406	-484	-760	541	2690		2507	4637	927	4935	Ok	
O2uc	555	2406	-555	-760	621	2690		4516	4637	1855	4935	Ok	
O3uc	524	2406	-524	-760	586	2690		4349	4637	1163	4935	Ok	
O4uc	515	2406	-515	-760	576	2690		2674	4637	1619	4935	Ok	
D1c	33	1203	-22	-760	37	1345		230	2110	1371	2110	Ok	
D2c	42	1203	-42	-760	47	1345		1949	2110	245	2110	Ok	
D3c	15	1203	-9	-760	17	1345		1769	2110	1126	2110	Ok	
O1c	470	2406	-470	-760	525	2690		2501	4637	787	4935	Ok	
O2c	544	2406	-544	-760	609	2690		4629	4637	1828	4935	Ok	
O3c	511	2406	-511	-760	572	2690		4450	4637	1032	4935	Ok	
O4c	503	2406	-503	-760	562	2690		2680	4637	1583	4935	Ok	
Max RATIO		0.231		0.731		0.231		0.998		0.650			

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	83.235	355.000	Ok
Tubesheet thickness flanged extension :	0.000	64.000	Ok
Tube thickness :	0.494	2.100	Ok
Tube-tubesheet fillet weld leg :	1.276	1.300	Ok
Tube-tubesheet groove weld leg :	1.276	1.300	Ok

Min Shell length of thk, (28.000) adj. to tubesheet: 418.437 mm

Min Channel length of thk, (20.000) adj. to tubesheet: 353.644 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +7.134 mm
 Due to Pressure Shell Compresses by : -0.142 mm
 Due to Pressure + Thermal Shell Extends by : +7.150 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	119.552	1.000	196.722	1.000
Tubesheet Shear Stress	263.405	1.000	290.861	1.000
Tube Tensile Stress	371.512	1.000	736.987	1.000
Tube Compressive Stress	312.083	1.000	154.767	1.000
Tube-Tubesheet Joint load	371.513	1.000	736.984	1.000
Shell Stress (Axial, Junction)	42.660	0.207	25.600	1.000
Tubesheet-Channel Junction	36.707	1.000	25.600	0.371
Tube Pressure Stress	290.929	1.000	190.123	1.000
Minimum MAWP	36.707		25.600	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	121.670	1.000	197.832	1.000
Tubesheet Shear Stress	265.929	1.000	395.171	1.000



Tube Tensile Stress	389.855	1.000	759.977	1.000
Tube Compressive Stress	432.921	1.000	539.925	1.000
Tube-Tubesheet Joint load	389.855	1.000	759.978	1.000
Shell Stress (Axial, Junction)	121.670	1.000	73.106	0.996
Tubesheet-Channel Junction	121.670	2.644	73.106	0.372
Tube Pressure Stress	290.929	1.000	194.704	1.000
Minimum MAPnc	121.670		73.106	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure UnCorroded	Due to Pressure Corroded	Due to Thermal Exp. UnCorroded	Due to Thermal Exp. Corroded	Computed Delta Ms UnCorroded	Computed Delta Ms Corroded
1	0.0162	0.0162	...	...	...	...
2	-0.1421	-0.1421	...	...	...	...
3	-0.1259	-0.1260	...	...	...	...
4	...	...	...	...	...	...
5	0.0162	0.0162	7.1314	7.1337	...	...
6	-0.1421	-0.1421	7.1314	7.1337	...	...
7	-0.1259	-0.1260	7.1314	7.1337	...	...
8	...	...	7.1314	7.1337	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness}))$$

$$= \max(355/4, \min(355, 28))$$

$$= 88.750 \text{ mm}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{ps}/\text{Shellside MAPnc}, 1)$$

$$= \min(25.5/73.1, 1)$$

$$= 0.349$$

Thickness Ratio = 0.35, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	29 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness}))$$

$$= \max(355/4, \min(355, 20))$$

$$= 88.750 \text{ mm}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{pt}/\text{Tubeside MAPnc}, 1)$$

$$= \min(10.2/122, 1)$$

$$= 0.084$$

Thickness Ratio = 0.084, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C